



SOLUTIONS OF NEWTONS LAWS OF MOTION

EXERCISE-1

PART - I

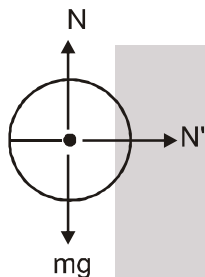
SECTION (A) :

A-1. Gravitational, Electromagnetic, Nuclear.

A-2. Newton's IIIrd Law

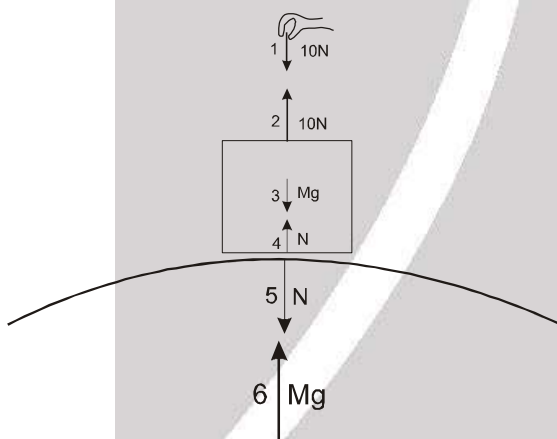
A-3. Newton's IInd Law

A-4.



Vertical wall does not exert force on sphere ($N' = 0$).

A-5.



For block
 $mg = 10 + N$ [Equilibrium]

$$\Rightarrow 1 \times 10 = 10 + N$$

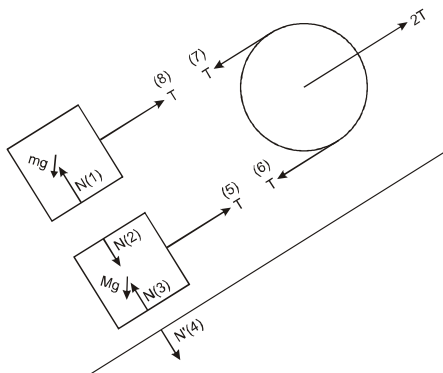
$$\Rightarrow N = 0$$

(1) and (2) are action reaction pair

(3) and (6) are action reaction pair

(4) and (5) are action reaction pair

A-6.

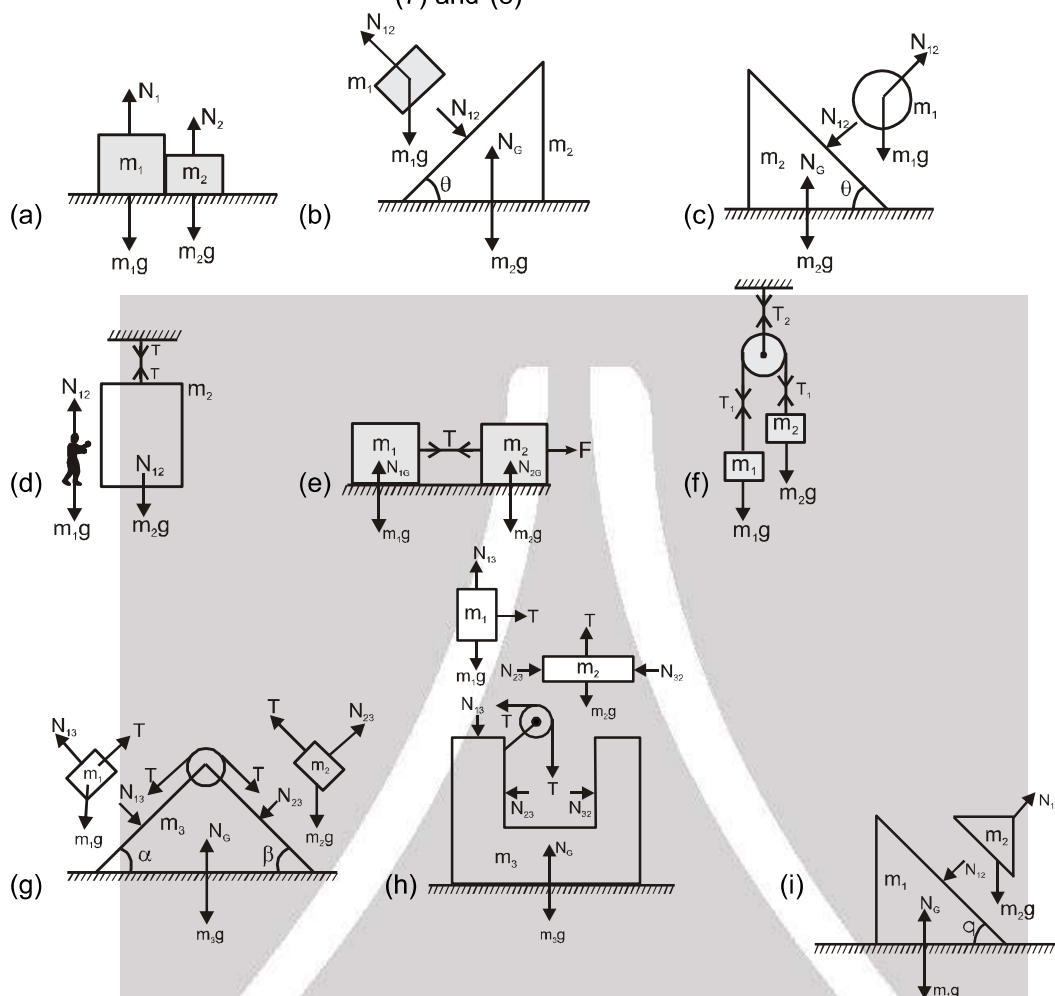




Action reaction pairs

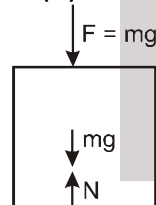
- (1) and (2)
(3) and (4)
(5) and (6)
(7) and (8)

A-7.



SECTION (B)

B-1.



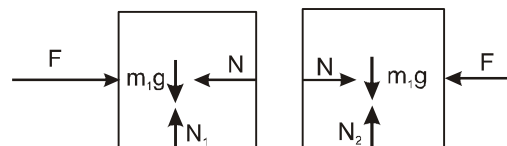
$$N = F + mg$$

[equilibrium]

$$\Rightarrow N = mg + mg$$

$$\Rightarrow N = 2mg$$

B-2.



It is obvious that both blocks will have same acceleration. If we take both block as one system then.

$$F - F = (m_1 + m_2) a$$

[Newton's second law in horizontal direction]

$$\Rightarrow a = 0$$





Now take m_1 as a system

[Newton's second law
in horizontal direction]

$$F - N = m_1 a$$

$$\Rightarrow F - N = 0$$

$$\Rightarrow F = N$$

$$m_1 g - N_1 = 0 \quad [\text{Equilibrium in vertical direction}]$$

Now take m_2 as system

$$N - F = m_2 a \quad [\text{Newton's second law
in horizontal direction}]$$

$$\Rightarrow N - F = 0$$

$$N = F$$

$$m_2 g - N_2 = 0 \quad [\text{Equilibrium in vertical direction}]$$

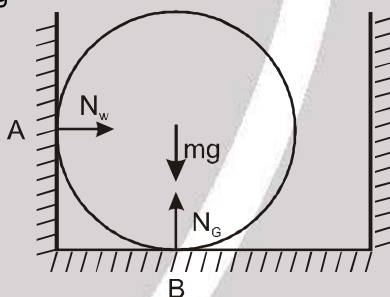
$$\Rightarrow N_2 = m_2 g$$

B-3. The sphere is in contact at two surfaces one at wall and one at ground. So one Normal reaction can be exerted at A and another at B.

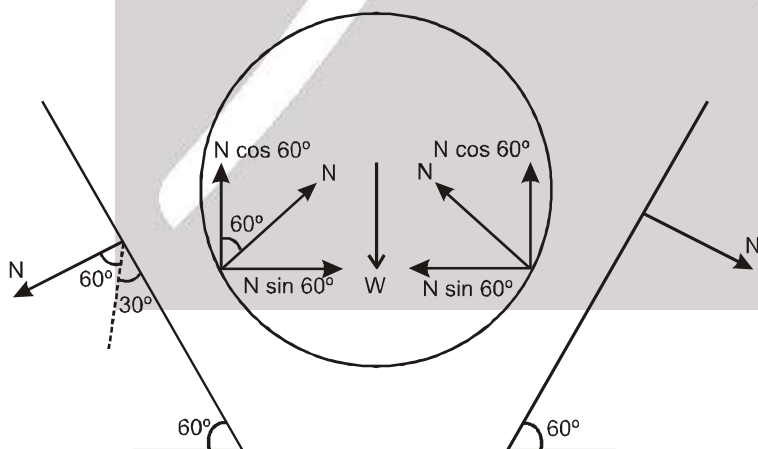
$$N_w = 0 \quad [\text{Equilibrium in horizontal direction}]$$

$$N_G - mg = 0 \quad [\text{Equilibrium in vertical direction}]$$

$$\Rightarrow N_G = mg$$



B-4.



Due to symmetry normal reactions due to left and right wall are same in magnitude
 $W - N \cos 60 - N \cos 60 = 0 \quad [\text{Equilibrium in vertical direction}]$

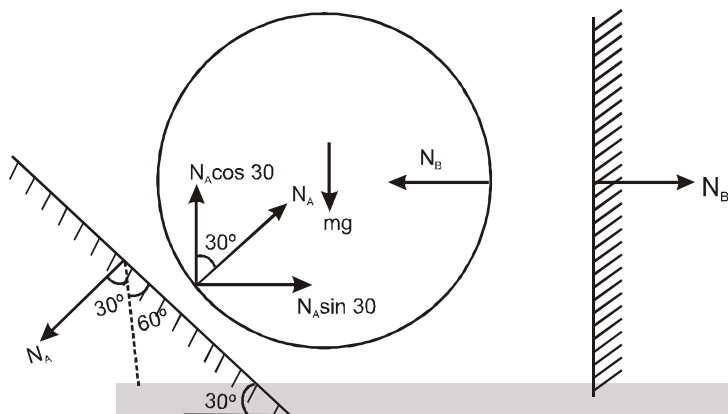
$$\Rightarrow W - \frac{N}{2} - \frac{N}{2} = 0$$

$$\Rightarrow N = W$$





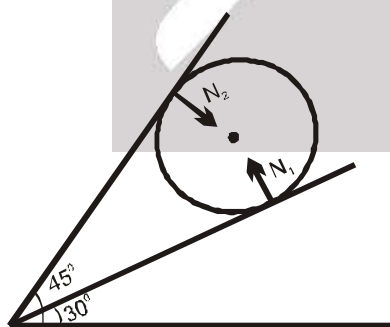
B-5.



$$\begin{aligned}
 &mg - N_A \cos 30 = 0 \quad \text{[Equilibrium in vertical direction]} \\
 \Rightarrow &N_A = \frac{mg}{\cos 30} \\
 \Rightarrow &N_A = \frac{1000}{\sqrt{3}} \text{ N} \\
 &N_B - N_A \sin 30 = 0 \quad \text{[Equilibrium in horizontal]} \\
 \Rightarrow &N_B = N_A \sin 30 \\
 \Rightarrow &N_B = \frac{1000}{\sqrt{3}} \cdot \frac{1}{2} \\
 \Rightarrow &N_B = \frac{500}{\sqrt{3}} \text{ N}
 \end{aligned}$$

B-6. $N_1 \cos 30^\circ = 50 + \frac{N_2}{\sqrt{2}}$

$$N_1 \frac{\sqrt{3}}{2} - \frac{N_2}{\sqrt{2}} = 50 \quad \dots\dots\dots (1)$$



$$N_1 \sin 30^\circ = \frac{N_2}{\sqrt{2}}$$

$$N_1 = \sqrt{2} N_2 \quad \dots\dots\dots (2)$$

Solving equation (1) & (2)

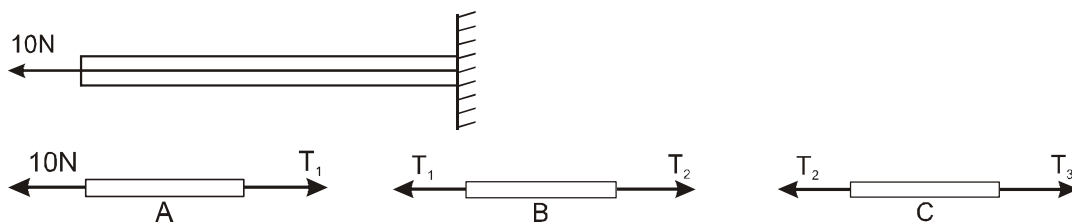
$$N_1 = 136.6 \text{ N}$$

$$N_2 = 96.59 \text{ N}$$



SECTION (C) :

C-1.



$$T_1 - 10 = 0$$

[Equilibrium of A]

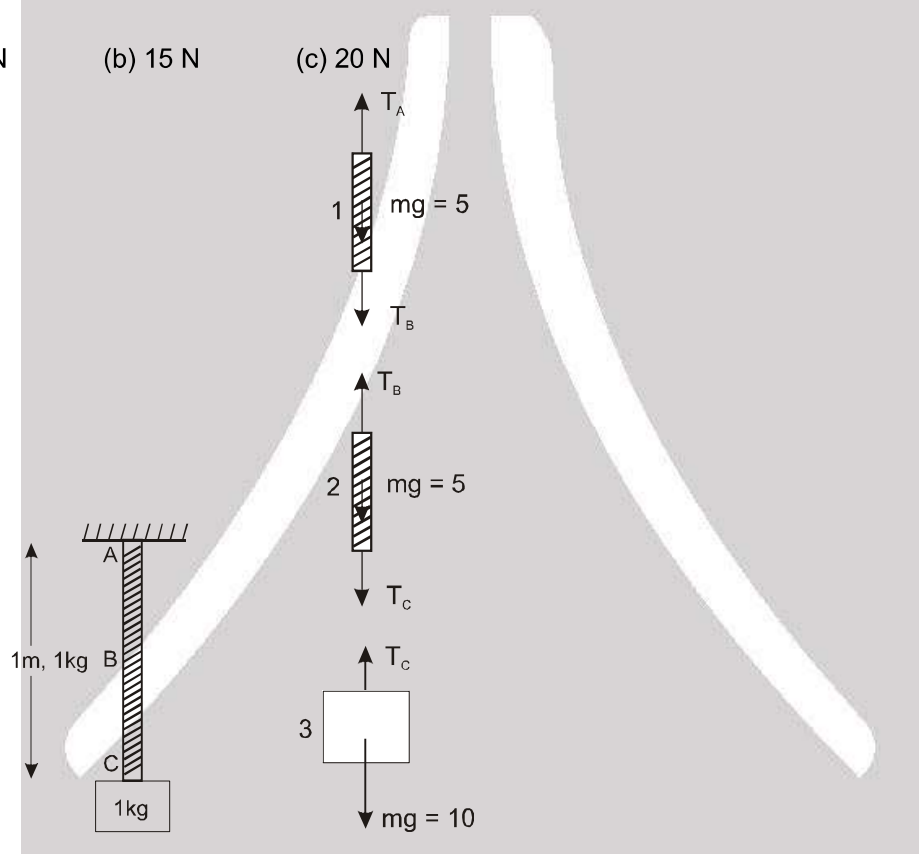
$$T_1 = 10 \text{ N}$$

Similarly T_2 and T_3 are also 10 N

C-2. (a) 10 N

(b) 15 N

(c) 20 N



$$T_C - 10 = 0 \quad \text{[Equilibrium of block]}$$

$$\Rightarrow T_C = 10 \text{ N}$$

$$T_B - T_C - 5 = 0 \quad \text{[Equilibrium of 2]}$$

$$\Rightarrow T_B - 10 - 5 = 0$$

$$\Rightarrow T_B = 15 \text{ N}$$

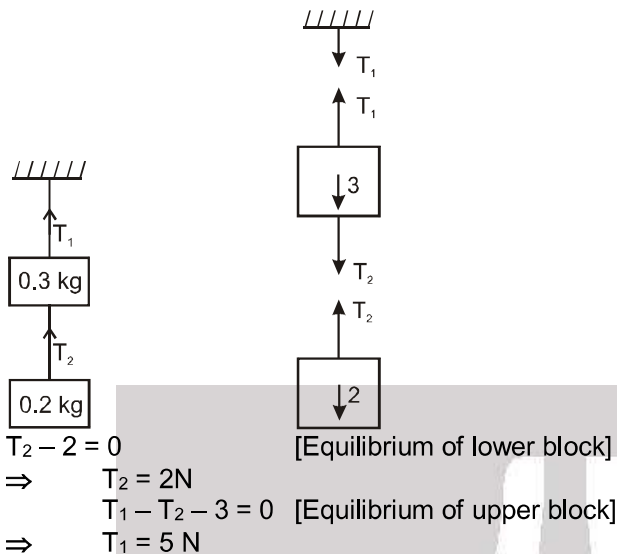
$$T_A - T_B - 5 = 0 \quad \text{[Equilibrium of 1]}$$

$$\Rightarrow T_A = 20 \text{ N}$$

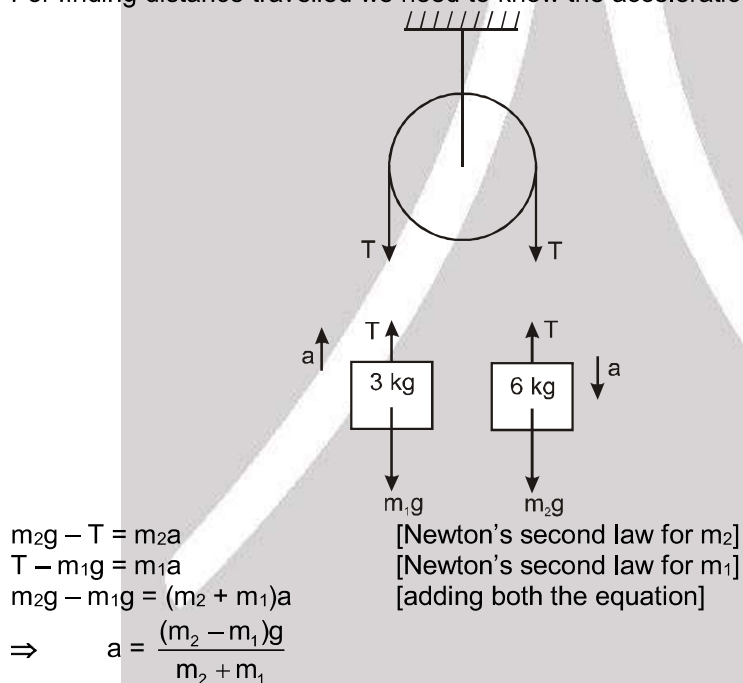




C-3.



C-4. For finding distance travelled we need to know the acceleration and initial velocity of block.



$$a = \frac{6 - 3}{6 + 3} \times g$$

$$a = \frac{g}{3} = \frac{10}{3} \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$= 0 \times 2 + \frac{1}{2} \times \frac{10}{3} \times 2^2$$

$$S = \frac{20}{3} \text{ m}$$

$$T - m_1g = m_1a$$

$$T = m_1 \left(g + \frac{g}{3} \right) = 3 \times \frac{40}{3} \quad T = 40 \text{ N}$$

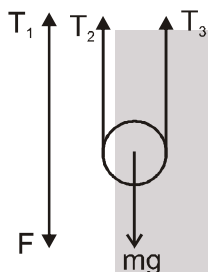
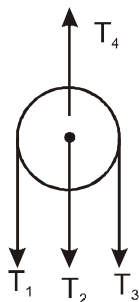
 Force exerted by clamp on pulley is $2T$

$$\Rightarrow 2 \times 40 = 80 \text{ N}$$





C-5.



$$T_1 = F \quad [\text{Equilibrium of string}]$$

$$T_3 = T_1 \quad [\text{String is massless and pulley is friction less so tension must be same on both sides of string}]$$

$$\Rightarrow T_3 = F$$

$$\text{Similarly } T_2 = F$$

$$T_5 = T_2 + T_3 \quad [\text{Equilibrium of lower pulley}]$$

$$\Rightarrow T_5 = 2F$$

$$T_5 = mg \quad [\text{Equilibrium of block}]$$

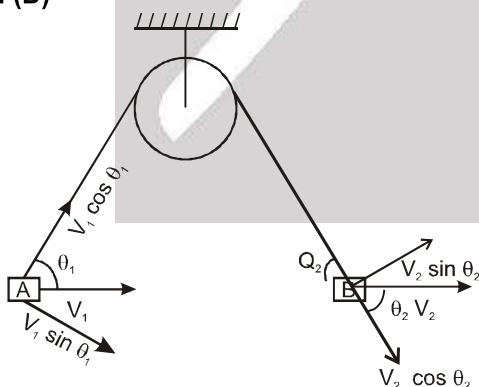
$$F = T_2 = T_3 = \frac{Mg}{2}$$

$$T_4 = T_1 + T_2 + T_3$$

$$[\text{Equilibrium of upper pulley}]$$

$$\Rightarrow T_4 = \frac{3Mg}{2}$$

Section (D)



D-1.

Since string is inextensible length of string can't change

$\therefore$ Rate of decreases of length of left string = rate of increase of length of right string

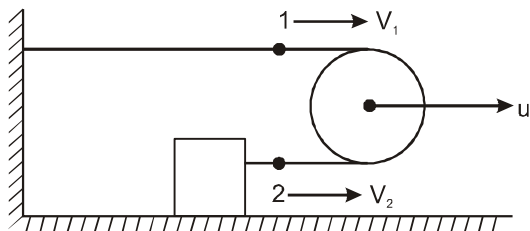
$$\Rightarrow V_1 \cos \theta_1 = V_2 \cos \theta_2$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{\cos \theta_2}{\cos \theta_1}$$





D-2.


 Velocity of point 1 is V_1 which is 0 because string is fixed.

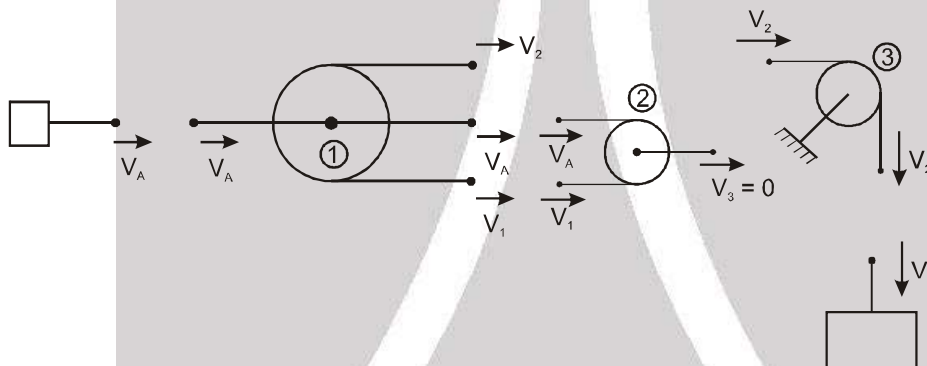
 Velocity of point 2 is V_2

$$\frac{V_1 + V_2}{2} = u$$

$$\frac{0 + V_2}{2} = u$$

$$V_2 = 2u$$

D-3.



$$V_A = \frac{V_1 + V_2}{2} \quad \text{[Pulley 1]}$$

$$V_1 + V_2 = 2V_A \quad \text{.....I}$$

$$\frac{V_A + V_1}{2} = V_3 \quad \text{[Pulley 2]}$$

$$V_A + V_1 = 0 \quad \text{.....II}$$

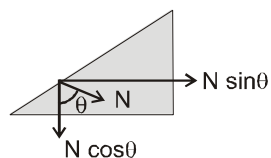
$$\Rightarrow V_1 = -0.6$$

$$-0.6 + V_2 = 2V_A$$

$$V_2 = 2 \times 0.6 + 0.6$$

$$V_2 = 1.8 \text{ m/s}$$

$$V_B = V_2 = 1.8 \text{ m/s}$$

 D-4. $\tan 37^\circ = \frac{a_A}{a_B}$ (wedge constrained relation)


$$N \sin 37^\circ = ma_B \quad \text{.....(i)}$$

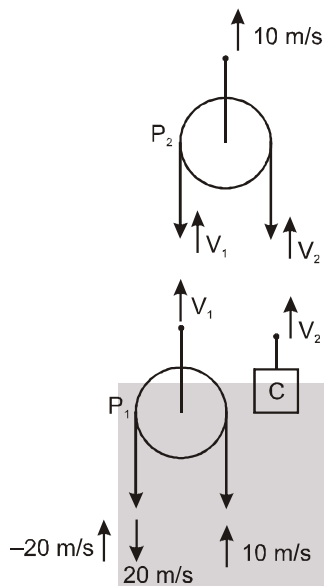
$$\text{For Rod} \rightarrow mg - N \cos 37^\circ = ma_A \quad \text{.....(ii)}$$

$$\text{From equation (1) \& (2) } a_A = \frac{9g}{25}, a_B = \frac{12g}{25}$$





D-5.



$$V_1 = \frac{10 - 20}{2}$$

[constrained relation of P_1]

$$V_1 = -5 \text{ m/s}$$

$$10 = \frac{-5 + V_2}{2}$$

$$V_2 = 25 \text{ m/s} \uparrow$$

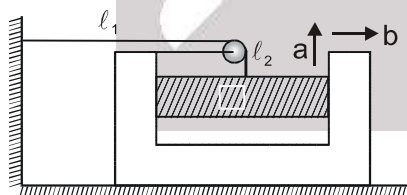
$$V_C = V_2 = 25 \text{ m/s upwards}$$

$$V_{P_1} = V_1 = -5 \text{ m/s}$$

$$\Rightarrow V_P = 5 \text{ m/s downward}$$

[because we have assumed upward direction as +ve for V_1]

D-6.



$$l_1 + l_2 = C$$

$$l_1'' + l_2 = 0$$

$$b - a = 0 \quad a = b$$

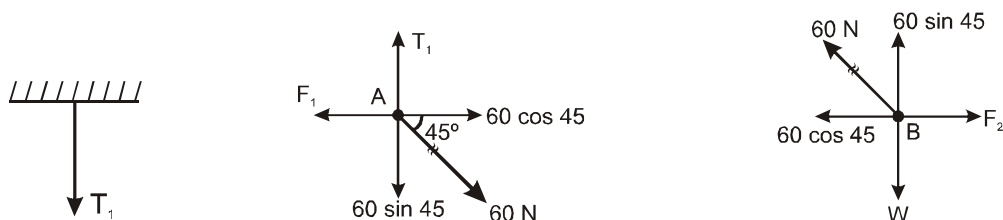
$$\text{Acceleration of A } b \hat{i} + b \hat{j}$$





SECTION (E)

E-1.



Since point A is massless net force on it must be zero other wise it will have ∞ acceleration.

$$\Rightarrow F_1 - 60 \cos 45 = 0$$

$$\Rightarrow F_1 = 30 \sqrt{2} \text{ N}$$

$$F_2 - 60 \cos 45 = 0$$

$$F_2 = 30 \sqrt{2} \text{ N}$$

$$W - 60 \sin 45 = 0$$

$$W = 30 \sqrt{2} \text{ N}$$

 E-2. $\vec{F} = m\vec{a}$

$$\vec{a} = a_x \hat{i} + a_y \hat{j}$$

$$= \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j}$$

$$= (10) \hat{i} + (18t) \hat{j}$$

at $t = 2 \text{ sec}$ $t = 2 \text{ sec}$

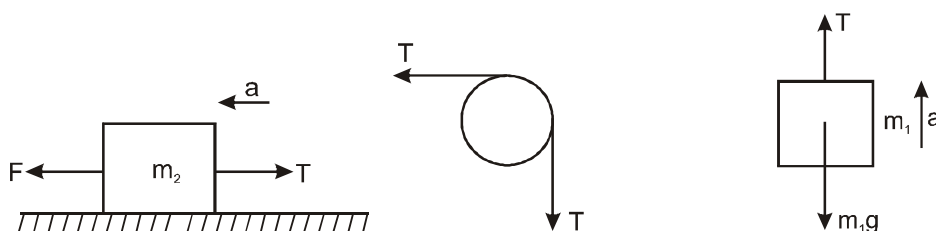
$$\vec{a} = 10 \hat{i} + 36 \hat{j}$$

$$\vec{F} = 3 (10 \hat{i} + 36 \hat{j})$$

$$= 30 \hat{i} + 108 \hat{j}$$

$$|\vec{F}| = \sqrt{30^2 + 108^2} = 112.08 \text{ N}$$

E-3.



It is obvious that acceleration of both the blocks is same in magnitude.





$$F - T = m_2 a \quad [\text{Newton's second law for } m_2]$$

$$T - m_1 g = m_1 a \quad [\text{Newton's second law for } m_1]$$

After adding the above equations.

$$F - m_1 g = (m_2 + m_1) a$$

$$\frac{m_1 g}{2} - m_1 g = (m_2 + m_1) a$$

$$\Rightarrow a = - \frac{m_1 g}{2(m_1 + m_2)}$$

The value of a is $-ve$ it means

$$a = \frac{m_1 g}{2(m_1 + m_2)} \text{ in the direction opposite to assumed direction}$$

E-4.

$$R_4 - mg = ma$$

$$R_4 - 1 = 0.1 \times 2$$

$$R_4 = 1.2 \text{ N}$$

$$R_3 - mg - R_4 = ma$$

$$R_3 - 1 - 1.2 = 0.1 \times 2$$

$$\Rightarrow R_3 = 2.4 \text{ N}$$

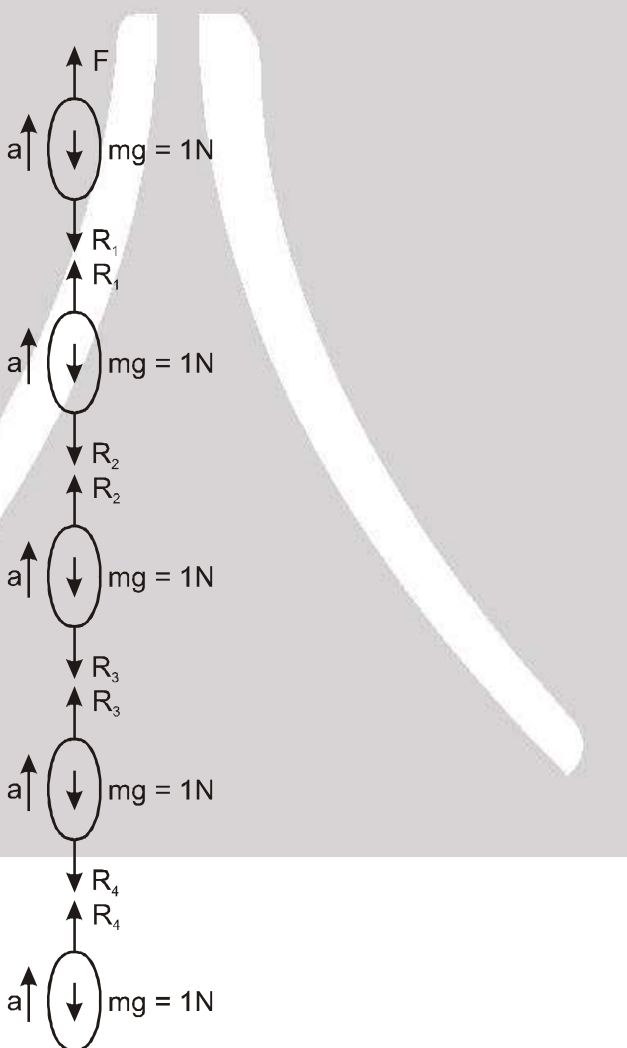
Similarly

$$R_2 = 3.6 \text{ N}$$

$$R_1 = 4.8 \text{ N}$$

$$F = 6 \text{ N}$$

$$\begin{aligned} F_{\text{net}} &= ma \\ &= 0.1 \times 2 \\ &= 0.2 \text{ N} \end{aligned}$$





E-5. $\int dp = p_f - p_i = \int F dt = \text{Area under the curve.}$

$p_i = 0$

Net Area $16 - 2 - 1 = 13 \text{ N-s}$

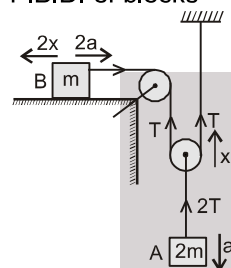
$V_f = 13/2 = 6.5 \text{ m/s}$

[As momentum is positive, particle is moving along positive x axis.]

E-6. (a) When the block m is pulled $2x$ towards left the pulley rises vertically up by x amount.

$\therefore a_B = 2a_A$

F.B.D. of blocks



$T = m2a$

F.B.D.

FBD of A,



$2mg - 2T = 2ma$

$mg - T = ma$

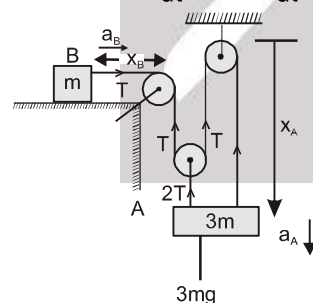
(1) + (2) $\Rightarrow mg = 3ma$

$a = g/3$

$\therefore a_B = 2g/3$

(b) $\ell = x_B + 3x_A$

$\Rightarrow 0 = \frac{d^2 x_B}{dt^2} + 3 \frac{d^2 x_A}{dt^2}$



$\Rightarrow 0 = -a_B + 3a_A$

$\Rightarrow a_B = 3a_A$

For B,

$T = ma_B$

For A,

$3mg - 3T = 3ma_A$

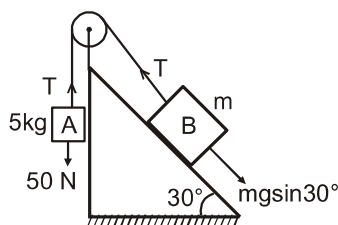
$mg - T = ma_A$

By (1), (2) & (3)

$\therefore a_B = 3g/4 \text{ Ans.}$



E-7. (a)
For Block (A)



$$T = 50 \text{ N} \quad \dots\dots(1)$$

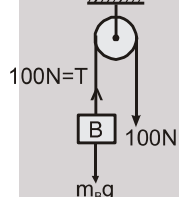
For Block (B)

$$T = mgsin30^\circ \quad \dots\dots(2)$$

$$\therefore 50 = m_B \times 10 \times 1/2$$

$$\Rightarrow m_B = 10 \text{ kg} \quad \text{Ans.}$$

(b)



$$T - m_B g = 0$$

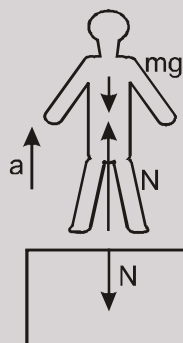
$$\Rightarrow 100 = m_B g$$

$$\therefore m_B = 10 \text{ kg} \quad \text{Ans.}$$

SECTION (F)

F-1. Reading of weighing machine is equal to the normal reaction Normal reaction is not affected by velocity of lift, it is only affected by acceleration of lift.

For I, II and III $a = 0$



$$N - mg = 0$$

$$N = mg = 600 \text{ N}$$

[Equilibrium of man]

For IV, VI and VII $IV, a = +2 \text{ m/s}^2$

$$N - mg = ma$$

[Newtons II law]

$$N = 60 \times 2 + 60 \times 10 = 720 \text{ N}$$

For V and VIII

$$a = -2 \text{ m/s}^2$$

$$N - mg = ma$$

[Newtons II law]

$$N = 60 \times (-2) + 60 \times 10 = 480 \text{ N}$$





F-2. Reading of spring balance is equal to the tension in spring balance which doesn't depend on velocity of lift but depend on acceleration.

For I, II and III $a = 0$ $a = 0$

$$T - 100 = 0 \quad [\text{Equilibrium}]$$

$$T = 100 \text{ N}$$

For IV, VI and VII

$$T - 100 = ma \quad [\text{Newton's II law}]$$

$$T - 100 = 10 \times 2$$

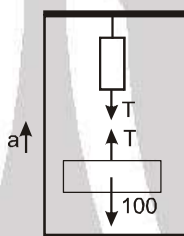
$$T = 120 \text{ N}$$

For V and VII

$$T - 100 = ma \quad [\text{Newton's II law}]$$

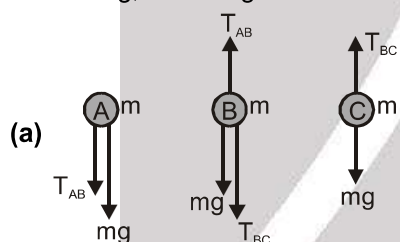
$$T - 100 = 10(-2)$$

$$T = 80 \text{ N}$$



F-3. Initially

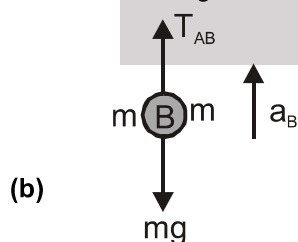
$$T_{AB} = 2mg, T_{BC} = mg$$



For A $2mg + mg = ma_A \Rightarrow a_A = 3g$

For B $T_{AB} - mg - T_{BC} = ma_B$
 $\Rightarrow 2mg - mg - mg = ma_B \Rightarrow ma_B = a_B = 0$

$T_{BC} - mg = ma_C \Rightarrow a_C = 0.$



$T_{AB} = 2mg$
 $T_{AB} - mg = ma_B$
 $2mg - mg = ma_B$
 $\Rightarrow a_B = g (\uparrow)$
 $a_A = 0 \text{ \& } a_C = g (\downarrow).$





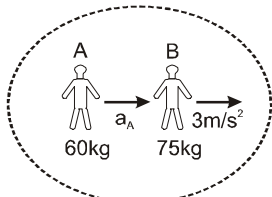
SECTION (G)

G-1. If we take both A and B as a system then there is no external force on system.

$$\Rightarrow m_A a_A + m_B a_B = 0 \quad [\text{Newton's II law for system}]$$

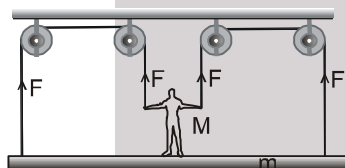
$$60 a_A + 75 \times 3 = 0$$

$$a_A = \frac{-15}{4} \text{ m/s}^2$$



–ve sign means that acceleration is in direction opposite to the assumed direction.

G-2.



$$4F - (M + m)g = (M + m)a$$

$$a = \frac{4F - (M + m)g}{M + m} = \frac{4F}{M + m} - g$$

G-3.

$$T_D = W_{A_{\text{app}}} + W_{B_{\text{app}}} + W_{C_{\text{app}}}$$

$$T_D = W_{A_{\text{आपसी}}} + W_{B_{\text{आपसी}}} + W_{C_{\text{आपसी}}}$$

$$= 10(10 - 2) + (15 \times 10) + 8(10 + 1.5)$$

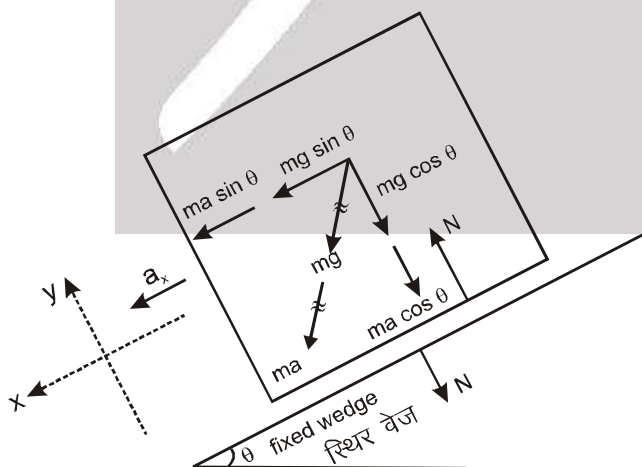
$$= 322 \text{ N Ans.}$$

SECTION (H)

H-1. Pseudo force depends on mass of object and acceleration of observer (frame) which is zero in this problem.

$\Rightarrow$ Pseudo force is zero.

H-2.



F.B.D. in frame of lift

It is obvious that block can accelerate only in x direction. ma is Pseudo force.

$$\Rightarrow mg \sin \theta + ma \sin \theta = ma_x \quad [\text{Newton's II law for block in x direction}]$$

$$\Rightarrow a_x = (g + a) \sin \theta$$





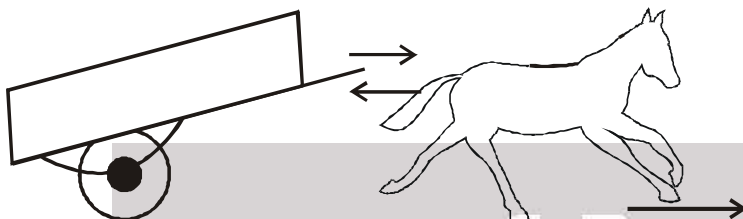
PART - II

SECTION (A)

A-1. Force exerted by string is always along the string and of pull type.

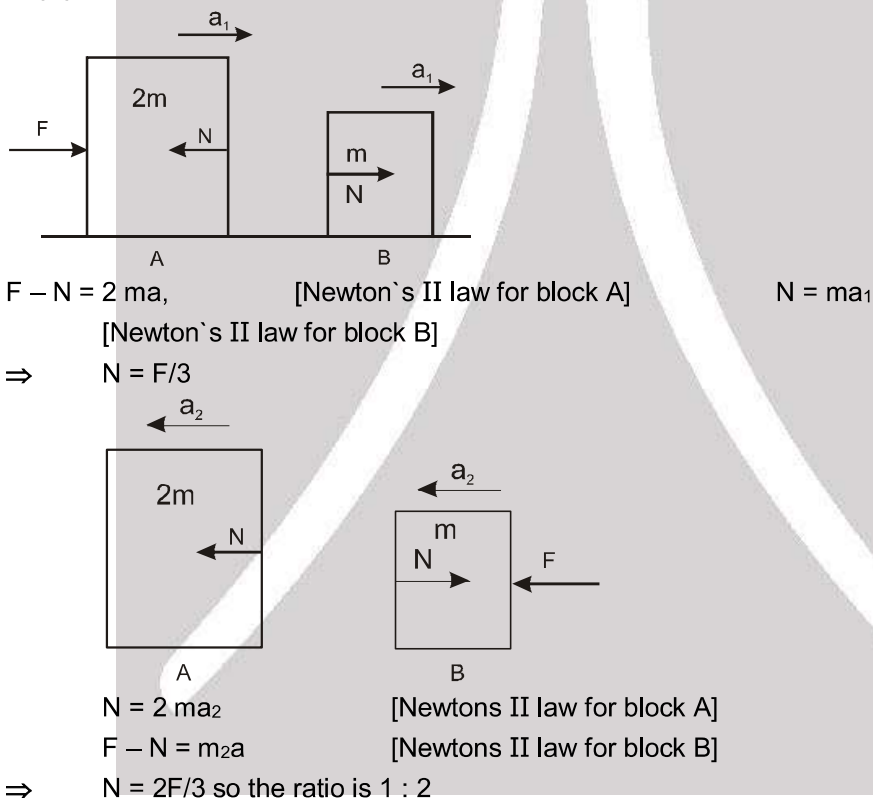
When there is a contact between a point and a surface the normal reaction is perpendicular to the surface and of push type.

A-2. The ground on the horse

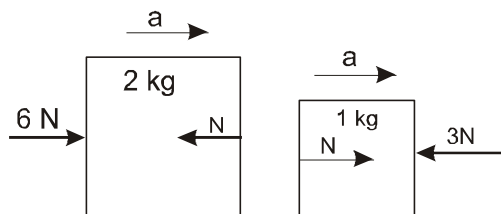


SECTION (B)

B-1.



B-2.



Both blocks are constrained to move with same acceleration.

$6 - N = 2a$ [Newton's II law for 2 kg block]

$N - 3 = 1a$ [Newton's II law for 1 kg block]

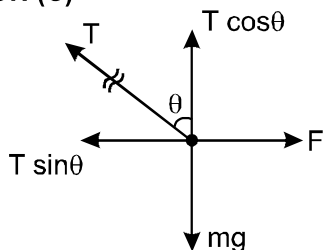
$\Rightarrow N = 4 \text{ Newton}$





SECTION (C)

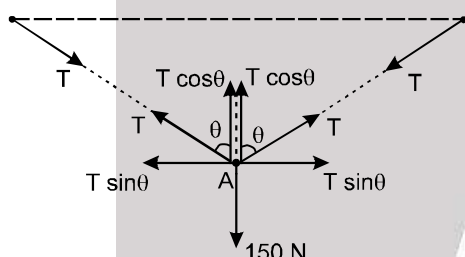
C-1.


 Point A is mass less so net force on it must be zero otherwise it will have ∞ acceleration.

$$\Rightarrow F - T \sin \theta = 0 \quad [\text{Equilibrium of A in horizontal direction}]$$

$$\Rightarrow T = \frac{F}{\sin \theta}$$

C-2.



$$T \cos \theta + T \cos \theta - 150 = 0 \quad [\text{Equilibrium of point A}]$$

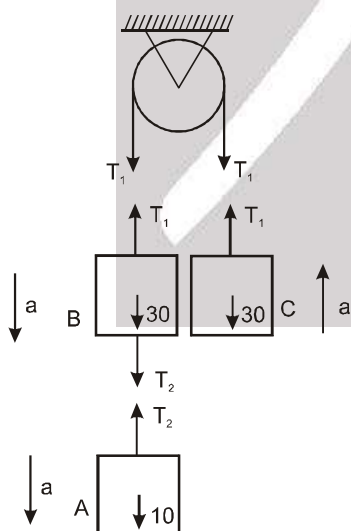
$$2 T \cos \theta = 150$$

$$T = \frac{75}{\cos \theta}$$

 When string become straight θ becomes 90°

$$\Rightarrow T = \infty$$

C-3.



$$10 - T_2 = 1 a \quad [\text{Newton's II law for A}]$$

$$T_2 + 30 - T_1 = 3 a \quad [\text{Newton's II law for B}]$$

$$T_1 - 30 = 3 a \quad [\text{Newton's II law for C}]$$

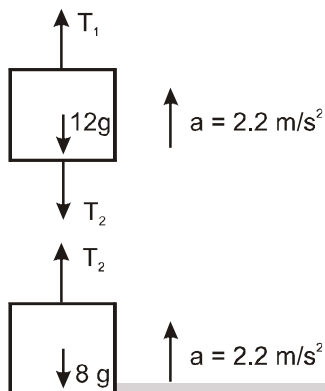
$$\Rightarrow a = g/7$$

$$\Rightarrow T_2 = 6g/7$$



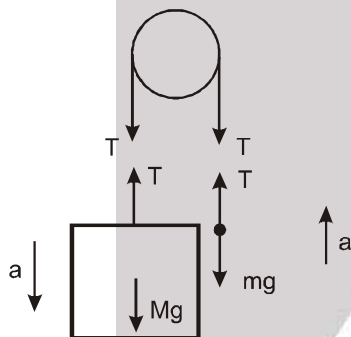


C-4.



$$\begin{aligned}
 &T_2 - 8g = 8a \quad [\text{Newton's II law for 8 kg block}] \\
 \Rightarrow &T_2 = 8 \times 2.2 + 8 \times 9.8 \\
 &= 96 \text{ N} \\
 &T_1 - 12g - T_2 = 12a \quad [\text{Newton's II law for 12 kg block}] \\
 \Rightarrow &T_1 = 12 \times 2.2 + 12 \times 9.8 + 96 \\
 &T_1 = 240 \text{ N}
 \end{aligned}$$

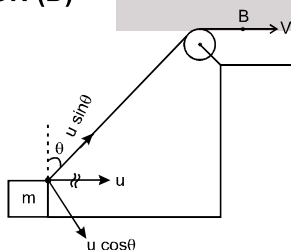
C-5.



$$\begin{aligned}
 &Mg - T = Ma \quad [\text{Newton's II law for M}] \\
 &T - mg = ma \quad [\text{Newton's II law for m}] \\
 \Rightarrow &T = \frac{2mMg}{m+M} \\
 \Rightarrow &T = 2mg \\
 &\text{Total downward force on pulley is } 2T = 4mg.
 \end{aligned}$$

SECTION (D)

D-1.



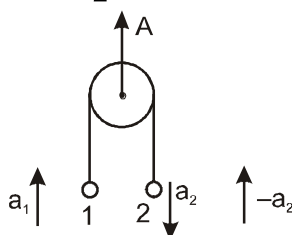
The length of string AB is constant.

$$\begin{aligned}
 \Rightarrow &\text{Speed A and B along the string are same } u \sin \theta = V \\
 &u \sin \theta = V \quad \quad \quad u = \frac{V}{\sin \theta}
 \end{aligned}$$

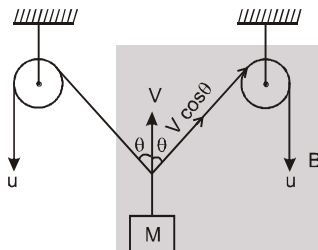




D-2. $A = \frac{a_1 - a_2}{2}$



D-3.



By symmetry we can conclude that block will move only in vertical direction.
Length of string AB remains constant
 $\therefore$ Velocity of point A and B along the string is same.

$$V \cos \theta = u \Rightarrow V = \frac{u}{\cos \theta}$$

D-4. Let $AB = \ell$, $B = (x, y)$

$$\vec{v}_B = v_x \hat{i} + v_y \hat{j}$$

$$\vec{v}_B = \sqrt{3} \hat{i} + v_y \hat{j} \rightarrow (i)$$

$$x^2 + y^2 = \ell^2$$

$$2x v_x + 2y v_y = 0 \Rightarrow \sqrt{3} + \frac{y}{x} v_y = 0$$

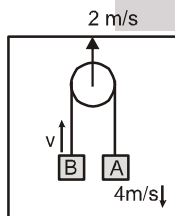
$$\Rightarrow \sqrt{3} + (\tan 60^\circ) v_y = 0 \Rightarrow v_y = -1$$

Hence from (i)

$$\vec{v}_B = \sqrt{3} \hat{i} - \hat{j}$$

Hence $v_B = 2 \text{ m/s}$

D-5.



V = (velocity of B w.r.t ground)

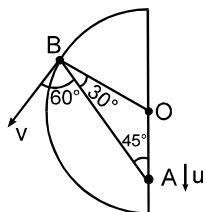
$$\frac{V - 4}{2} = 2 \quad V = 8 \text{ m/s (velocity of B w.r.t ground)}$$

$$V' = 6 \text{ m/s (velocity of B w.r.t lift)}$$





D-6. $u \cos 45^\circ = v \cos 60^\circ$



or $v = \sqrt{2} u$

SECTION (E)

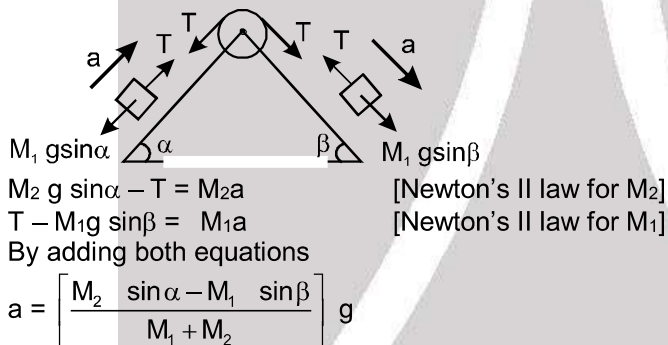
E-1. $\vec{F} = m\vec{a}$

$\vec{a} = \frac{d\vec{v}}{dt}$

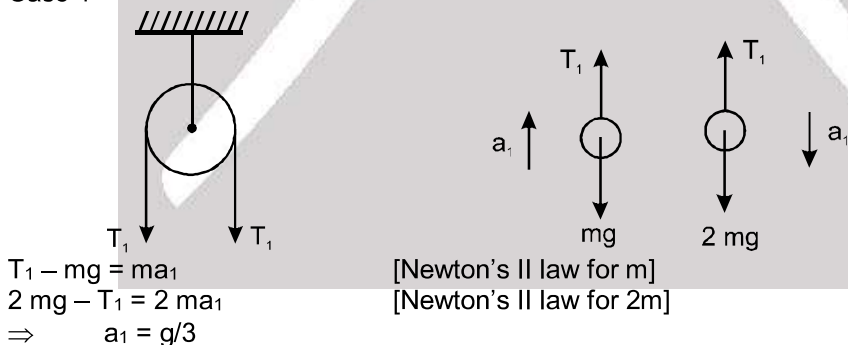
E-2. $\vec{F} = m\vec{a}$

E-3. In free fall gravitation force acts.

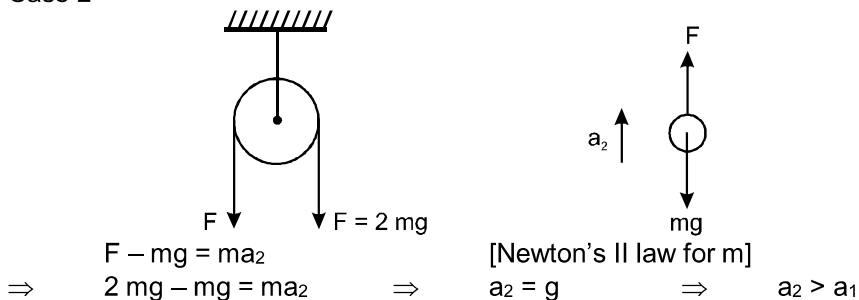
E-4.



E-5. Case 1



Case 2





E-6.

$F = m_1 4$
 $F = m_2 6$
 $F = (m_1 + m_2) a$

[Newton's II law for m_1]
 [Newton's II law for m_2]
 [Newton's II law for $(m_1 + m_2)$]

$$\Rightarrow F = \left[\frac{F}{4} + \frac{F}{6} \right] a \Rightarrow 1 = \left[\frac{1}{4} + \frac{1}{6} \right] a \Rightarrow a = 2.4 \text{ m/s}^2.$$

E-7.

Due to symmetry we can say net force on body M is 0.
 $\therefore$ Acceleration is 0

E-8. $mg - \frac{3}{4} mg = ma$ [Newton's II law for man]
 $\Rightarrow a = g/4$

E-9. $\vec{F} = 6 \hat{i} - 8 \hat{j} + 10 \hat{k}$
 $\vec{F} = m \vec{a}$
 $|\vec{F}| = m |\vec{a}|$
 $\sqrt{6^2 + 8^2 + 10^2} = m \cdot 1$ $m = 10\sqrt{2} \text{ kg}.$

E-10. $v^2 = u^2 + 2as$ $0^2 = 1^2 + 2 \frac{F}{m} x$
 $x = \frac{-m}{2F}$ $v^2 = u^2 + 2as$
 $0^2 = 3^2 + \frac{2F^1}{m} x$ $0 = 9 + \frac{2F^1}{m} \left(\frac{-m}{2F} \right) \Rightarrow F^1 = 9F$

E-11.

$Mg \sin \theta - T = Ma$ [Newton's II law for block 1]
 $T = Ma$ [Newton's II law for block 2]

By dividing both equations

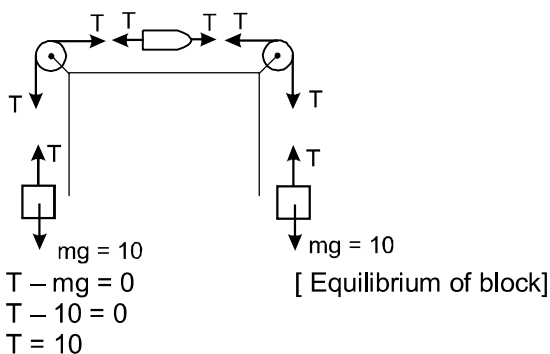
$$2T = Mg \sin \theta \quad T = \frac{Mg \sin \theta}{2}$$





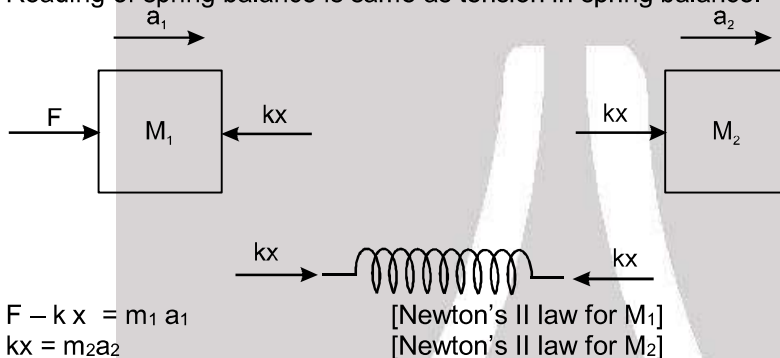
SECTION (F)

F-1.

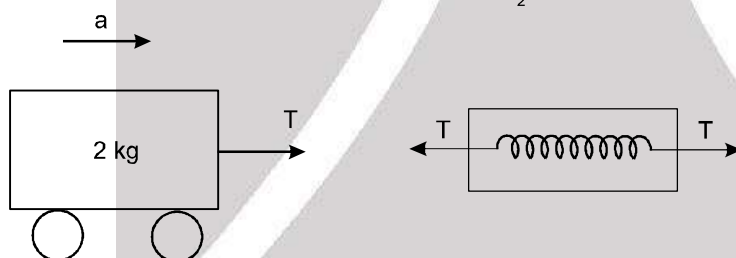


Reading of spring balance is same as tension in spring balance.

F-2.



F-3.



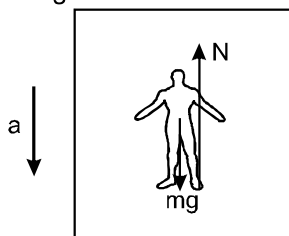
Reading of spring balance is same as tension in the balance.

$$\Rightarrow T = 10g = 98 \text{ N}$$

$$T = 2a$$

[Newton's II law for 2 kg block]

$$\Rightarrow a = 49 \text{ m/s}^2$$

 F-4. Weight of man in stationary lift is mg .


$$mg - N = ma \quad [\text{Newton's II law for man}]$$

$$\Rightarrow N = m(g - a)$$

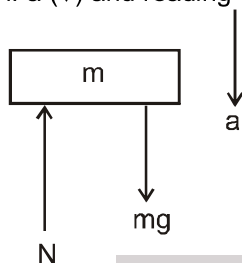
 Weight of man in moving lift is equal to N .

$$\Rightarrow \frac{mg}{m(g - a)} = \frac{3}{2} \Rightarrow a = \frac{g}{3}$$





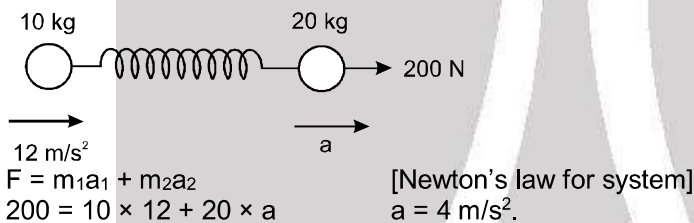
- F-5.** $N = m(g - a)$, $N < mg$ if a ($\downarrow$)
 $N = m(g + a)$, $N > mg$ if a ($\uparrow$)
 Reading of spring balance is less than m
 if a ($\downarrow$) and reading of spring balance is



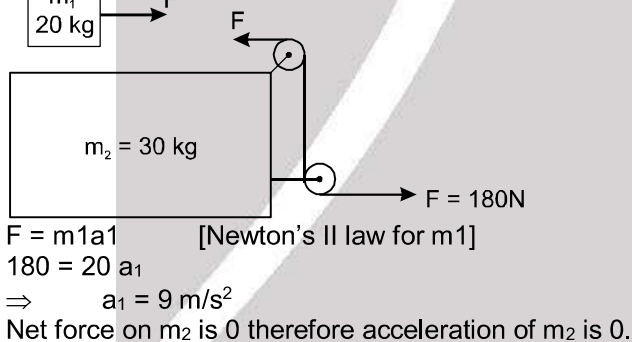
greater than m if a ($\uparrow$)

SECTION (G)

G-1.

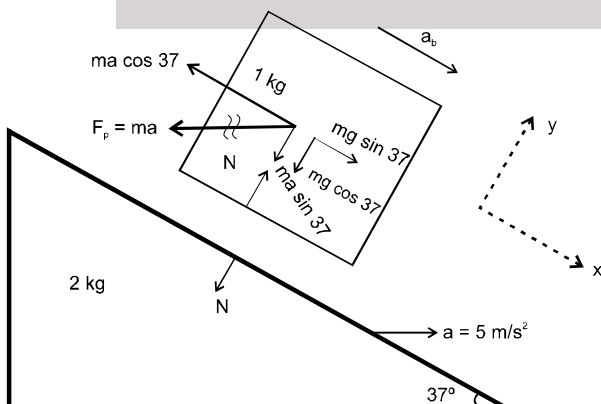


G-2.



SECTION (H)

H-1.



FBD of block is shown w.r.t. wedge and FBD of wedge is shown w.r.t. ground. F_P is pseudo force.

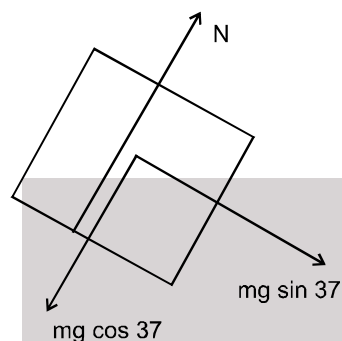
$$mg \sin 37 - ma \cos 37 = ma_b$$

$$\Rightarrow a_b = g \sin 37 - a \cos 37 = 10 \times 3/5 - 5 \times 4/5 = 2 \text{ m/s}^2 \text{ w.r.t. wedge}$$





- $\Rightarrow$ Block is not stationary w.r.t. wedge
 $N - ma \sin 37 - mg \cos 37 = 0$ [Newton's II law for block]
 $\Rightarrow N = 1 \times 10 \times 4/5 + 1 \times 5 \times 3/5$
 $\Rightarrow N = 11 \text{ N.}$
 Net force acting on block w.r.t. ground.

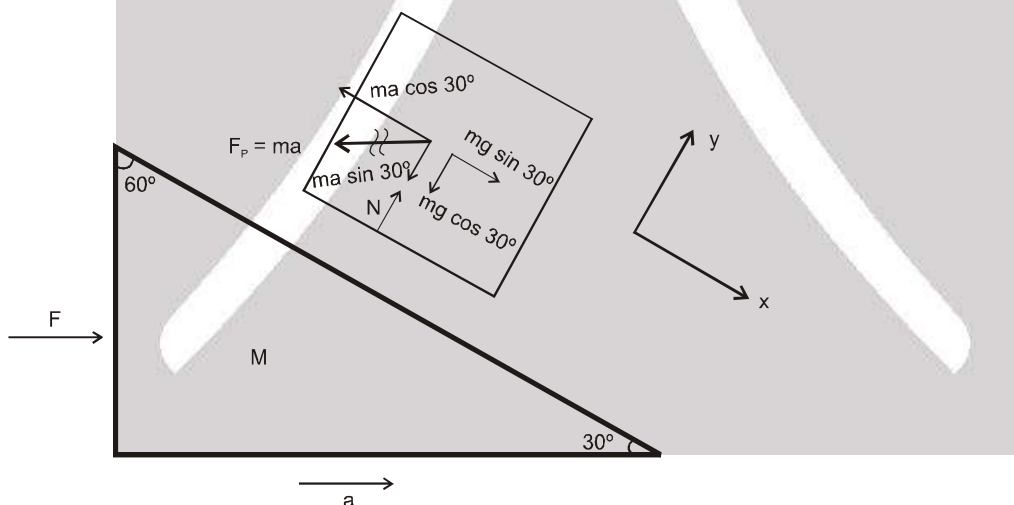


$$F = \sqrt{(mg \sin 37)^2 + (mg \cos 37 - N)^2}$$

$$= \sqrt{\left(10 \times \frac{3}{5}\right)^2 + \left(10 \times \frac{4}{5} - 11\right)^2} = \sqrt{6^2 + 3^2}$$

$$F = 3\sqrt{5} \text{ N.}$$

H-2.



F.B.D. of wedge is w.r.t. ground and

F.B.D. of block is w.r.t. wedge.

Let a is the acceleration of wedge due to force F .

F_P is pseudo force on block

$$mg \sin 30^\circ - ma \cos 30^\circ = 0 \quad [\text{Equilibrium of block in } x \text{ direction w.r.t. wedge}] \quad a = g \tan 30^\circ$$

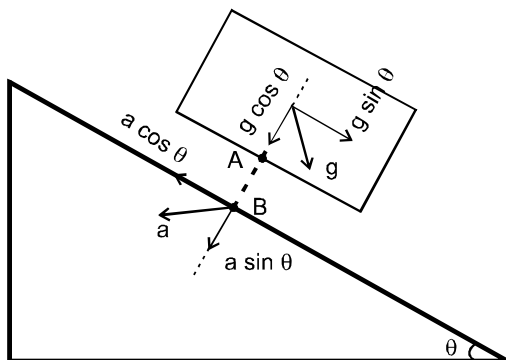
$$F = (M + m)a \quad [\text{Newton's II law for the system of block and wedge in horizontal direction}]$$

$$\Rightarrow F = (M + m)g \tan 30^\circ.$$





H-3.


 Acceleration of point A and B must be some along the line $\perp$ to the surface

$$\Rightarrow a \sin \theta = g \cos \theta$$

$$a = g \cot \theta$$

PART - III

1. Let a be acceleration of two block system towards right

$$\therefore a = \frac{F_2 - F_1}{m_1 + m_2}$$

 The F.B.D. of m_2 is

$$\therefore F_2 - T = m_2 a$$

$$\text{Solving } T = \frac{m_1 m_2}{m_1 + m_2} \left(\frac{F_2}{m_2} + \frac{F_1}{m_1} \right)$$

 (B) Replace F_1 by $-F_1$ is result of A

$$\therefore T = \frac{m_1 m_2}{m_1 + m_2} \left(\frac{F_2}{m_2} - \frac{F_1}{m_1} \right)$$

 (C) Let a be acceleration of two block system towards left

$$\therefore a = \frac{F_2 - F_1}{m_1 + m_2}$$

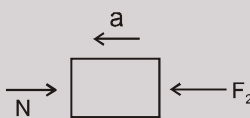
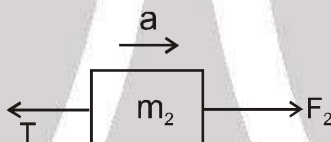
 The FBD of m_2 is

$$\therefore F_2 - N = m_2 a$$

$$\text{Solving } N = \frac{m_1 m_2}{m_1 + m_2} \left(\frac{F_1}{m_1} + \frac{F_2}{m_2} \right)$$

 (D) Replace F_1 by $-F_1$ in result of C

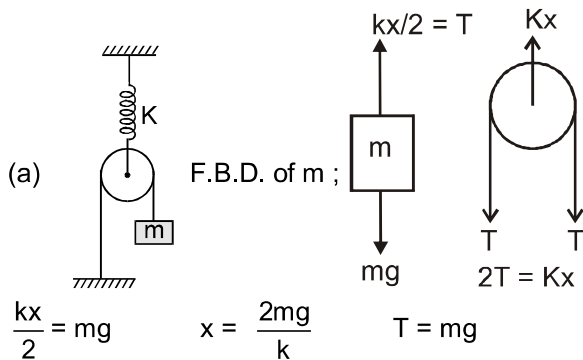
$$N = \frac{m_1 m_2}{m_1 + m_2} \left(\frac{F_2}{m_2} - \frac{F_1}{m_1} \right)$$



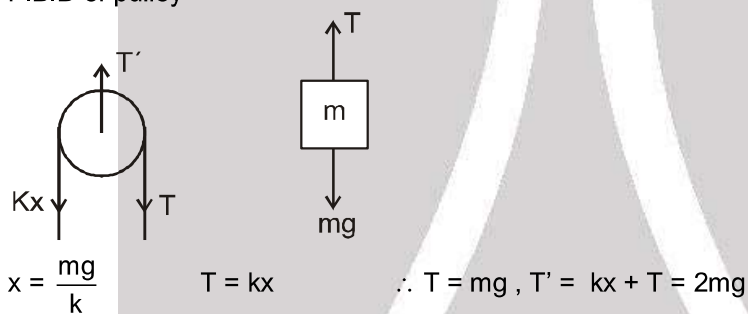
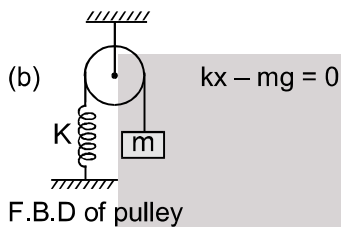


2.

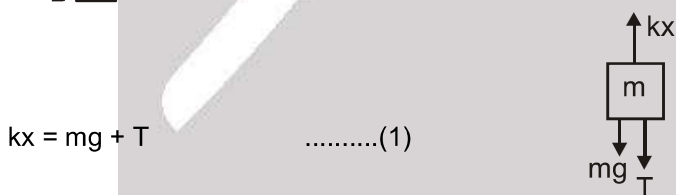
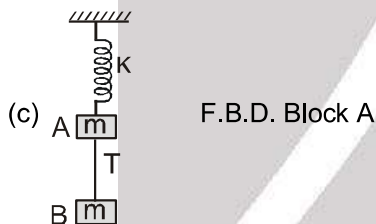
(a)



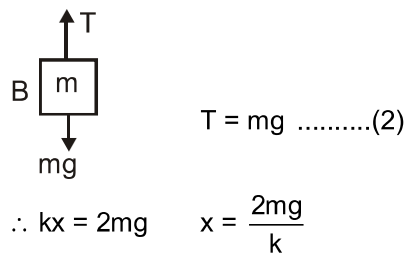
(b)

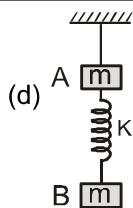


(c)

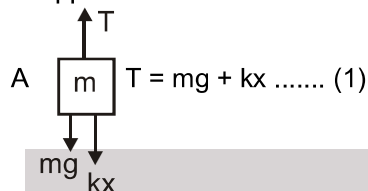


F.B.D. Block B





F.B.D. of Upper Block A



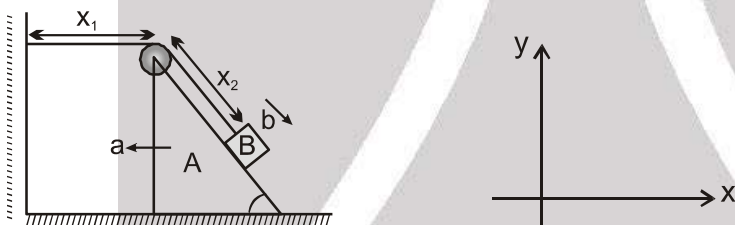
F.B.D. of Lower Block (B)



$$kx = mg \dots\dots (2) \therefore x = \frac{mg}{k}$$

By (1) & (2) $T = 2mg$

3.



(a) Let b be acceleration of block B w.r.t. wedge

i.e. $\vec{a}_{BW} = b \vec{a}_{BW} = b \cos \theta \hat{i} - b \sin \theta \hat{j}$

$$\ell = x_1 + x_2 \dots\dots(1)$$

$$\Rightarrow 0 = \frac{dx_1}{dt} + \frac{dx_2}{dt} \Rightarrow 0 = -a + b$$

$$\Rightarrow b = a \dots\dots(2)$$

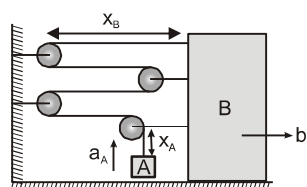
$$\therefore \vec{a}_{BW} = a \cos \theta \hat{i} - a \sin \theta \hat{j}$$

$$\vec{a}_{WG} = \text{acceleration of wedge w.r.t. ground} = -a \hat{i} \dots\dots(3)$$

$$\vec{a}_{BG} = \vec{a}_{BW} + \vec{a}_{WG}$$

$$\therefore \vec{a}_{BG} = (a \cos \theta - a) \hat{i} - a \sin \theta \hat{j} \text{ Ans.}$$

(b)

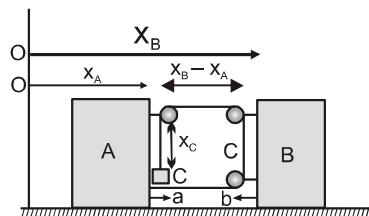


$$\ell = 4x_B + x_A \Rightarrow 0 = 4 \frac{d^2 x_B}{dt^2} + \frac{d^2 x_A}{dt^2} ; \quad \frac{d^2 x_A}{dt^2} = -a_{AB} ; \quad \frac{d^2 x_B}{dt^2} = b \Rightarrow 4b = a_{AB}$$





(c)



$$\vec{a}_{CA} = \frac{d^2 x_C}{dt^2} ; a = \frac{d^2 x_A}{dt^2} , b = -\frac{d^2 x_B}{dt^2}$$

$$\text{Length} = x_C + x_B - x_A + C + x_B - x_A$$

$$\Rightarrow \ell = x_C + 2x_B - 2x_A + C$$

$$\Rightarrow 0 = \frac{d^2 x_C}{dt^2} + 2 \frac{d^2 x_B}{dt^2} - 2 \frac{d^2 x_A}{dt^2}$$

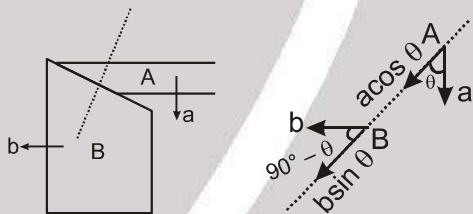
$$\Rightarrow 0 = a_{CA} - 2b - 2a \quad \therefore \vec{a}_{CA} = -(2a + 2b)$$

$$\vec{a}_{CG} = \vec{a}_{CA} + \vec{a}_{AG} = -(2a + 2b) \hat{j} + a \hat{i}$$

$$\therefore \vec{a}_{CG} = a \hat{i} - 2(a+b) \hat{j} \text{ Ans.}$$

 (d) Let a be acceleration of wedge A.

Acceleration of blocks A & B along normal to contact surface (shown by dotted line) must be equal.

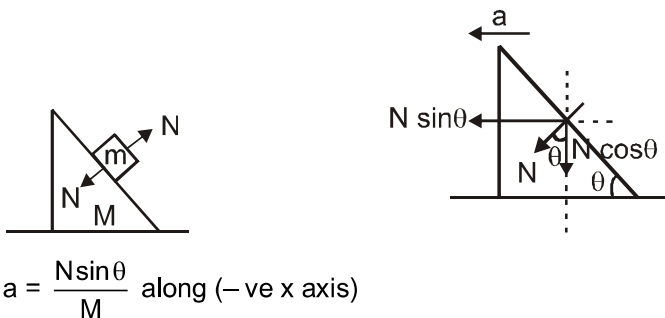


$$\text{i.e. } b \sin \theta = a \cos \theta \quad a = b \tan \theta$$

$$\therefore \vec{a}_A = -b \tan \theta \hat{j} \text{ Ans.}$$

EXERCISE-2 PART - I

1.



$$a = \frac{N \sin \theta}{M} \text{ along } (-ve \text{ x axis})$$





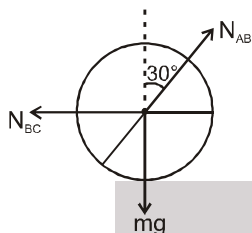
2. The free body diagram of cylinder is as shown.

Since net acceleration of cylinder is horizontal,

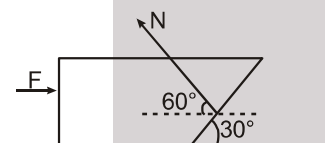
$$N_{AB} \cos 30^\circ = mg \quad \text{or} \quad N_{AB} = mg \quad \dots (1)$$

$$\text{and} \quad N_{BC} - N_{AB} \sin 30^\circ = ma \quad \text{or} \quad N_{BC} = ma + N_{AB} \sin 30^\circ \quad \dots (2)$$

Hence N_{AB} remains constant and N_{BC} increases with increase in a .



- 3.

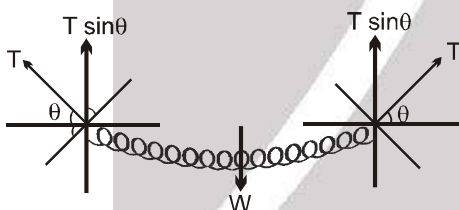


Acceleration of two mass system is $a = \frac{F}{2m}$ leftward

FBD of block A

$$N \cos 60^\circ - F = ma = \frac{mF}{2m} \quad \text{solving } N = 3F$$

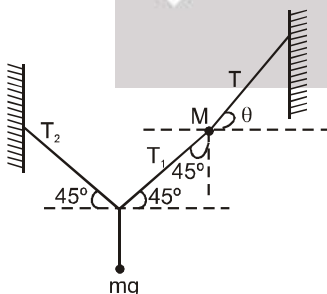
- 4.



$$2T \sin \theta = W$$

$$T = \frac{W}{2} \operatorname{cosec} \theta$$

- 5.



$$T_1 \cos 45^\circ = T_2 \cos 45^\circ$$

$$\Rightarrow T_1 = T_2$$

$$(T_1 + T_2) \sin 45^\circ = mg$$

$$\sqrt{2} T_1 = mg$$

$$T_1 = \frac{mg}{\sqrt{2}}$$





$$T \sin \theta = Mg + \frac{T_1}{\sqrt{2}}$$

$$T \sin \theta = Mg + \frac{mg}{2} \quad \dots\dots\dots(i)$$

$$T \cos \theta = \frac{T_1}{\sqrt{2}} = \frac{mg}{2} \quad \dots\dots\dots(ii)$$

Dividing (i) and (ii)

$$\tan \theta = \frac{M+m/2}{m/2} = 1 + \frac{2M}{m} \text{ Ans.}$$

6. $T = mg$

$$2T \cos \theta = T'$$

$$T' = Mg$$

$$2mg \cos \theta = Mg$$

$$\cos \theta = \frac{M}{2m} < 1$$

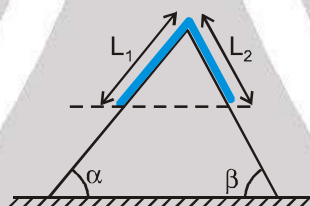
$$M < 2m$$

7. Let L_1 and L_2 be the portions (of length) of rope on left and right surface of wedge as shown

$\therefore$ Magnitude of acceleration of rope

$$a = \frac{\frac{M}{L} [L_1 \sin \alpha - L_2 \sin \beta] g}{M} = 0$$

$$(\because L_1 \sin \alpha = L_2 \sin \beta)$$



8. By setting string length constant

$$L = \ell_1 + 2\ell_2 + 2\ell_3$$

After differentiation $L' = 0$ so

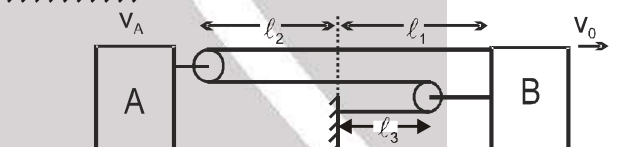
$$-2v_A + v_0 + 2v_0 = 0$$

$$\Rightarrow 3v_0 = 2v_A$$

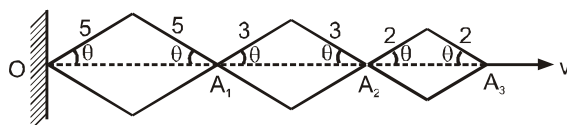
$$v_A = \frac{3}{2} v_0$$

$$v_{AB} = v_A - v_B$$

$$= \frac{v_0}{2} \text{ towards right.}$$



9.



$$x = 20 \cos \theta$$

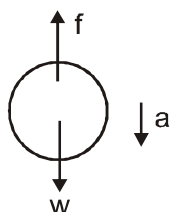
$$y = 16 \cos \theta$$

$$v = \frac{dx}{dt} = -20 \sin \theta \frac{d\theta}{dt} \Rightarrow u = \frac{dy}{dt} = -16 \sin \theta \frac{d\theta}{dt} \Rightarrow u = \frac{4}{5} v = 0.8 v$$





10.



$$w - f = ma \quad w - ma = g$$

$$w \left\{ 1 - \frac{m}{w} a \right\} = f \quad w \left\{ 1 - \frac{m}{mg} a \right\} = f$$

$$w \left\{ 1 - \frac{a}{g} \right\} = f$$

11.

$$\text{Length of groove} = \sqrt{3^2 + 4^2} = 5\text{m}$$

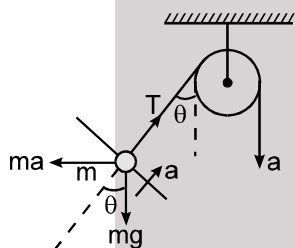
$$\text{Acceleration along the incline} = g \sin \theta = g \sin 30^\circ = g/2$$

$$\text{Acceleration along the groove} = g/2 \cos (90 - \alpha) = g/2 \sin \alpha = \frac{g}{2} \times \frac{4}{5} = 4\text{m/s}^2$$

$$v^2 = 2as$$

$$v = \sqrt{2 \times 4 \times 5} = \sqrt{40} \text{ m/sec.}$$

12.



(Force diagram in the frame of the car)

Applying Newton's law perpendicular to string

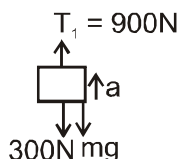
$$mg \sin \theta = ma \cos \theta$$

$$\tan \theta = \frac{a}{g}$$

Applying Newton's law along string

$$\Rightarrow T - m\sqrt{g^2 + a^2} = ma \quad T = m\sqrt{g^2 + a^2} + ma \text{ Ans.}$$

13.



$$900 - 300 - m \times 10 = ma \quad 600 = m(10 + a)$$

$$\frac{600}{10 + a} = m$$

$$\frac{600}{10 + 10} = m = \frac{600}{20} = 30 \text{ kg.}$$



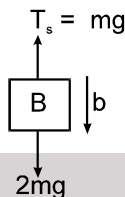


14. For first case tension in spring will be



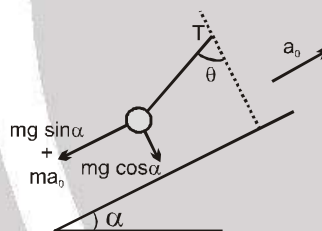
$T_s = 2mg$ just after 'A' is released.

$$2mg - mg = ma \Rightarrow a = g$$



In second case $T_s = mg$
 $2mg - mg = 2mb$
 $b = g/2$
 $a/b = 2$

15. $T \sin \theta = m(g \sin \alpha + a_0)$
 $T \cos \theta = mg \cos \alpha$
 $\Rightarrow \tan \theta = \left(\frac{g \sin \alpha + a_0}{g \cos \alpha} \right)$
 $\theta = \tan^{-1} \left(\frac{g \sin \alpha + a_0}{g \cos \alpha} \right)$



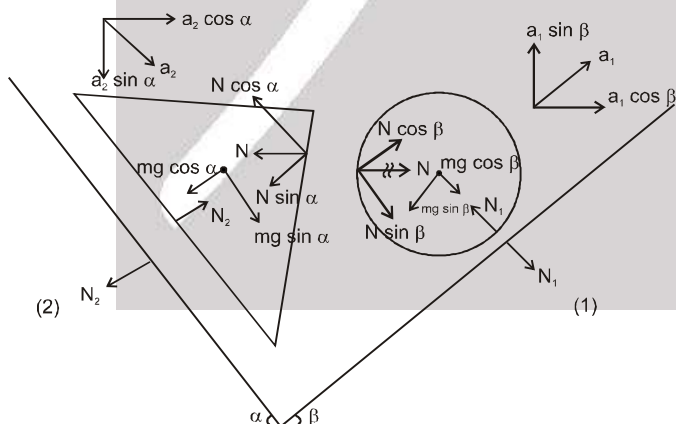
16. Slope of $v_{rel} - t$ curve is Constant.

$$\Rightarrow a_{rel} = \text{Const.} = a_1 - a_2 \neq 0$$

Inference that at least one reference frame is accelerating both can't be non - accelerating simultaneously.

PART - II

- 1.



It is obvious that acceleration of cylinder is $\parallel$ to the wedge 1 and acceleration of triangular block is $\parallel$ to the wedge 2.

$$a_2 \cos \alpha = a_1 \cos \beta$$

[constrained relation between the contact surface of block and cylinder]

$$N \cos \beta - m_1 g \sin \beta = m_1 a_1$$

[Newton's II law for cylinder along the direction parallel to the wedge 1]

$$m_2 g \sin \alpha - N \cos \alpha = m_2 a_2$$

[Newton's II law for block along the direction parallel to the wedge 2]

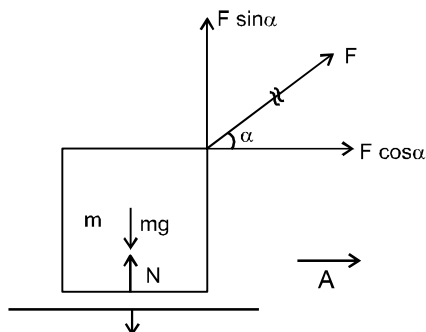
By solving equation I, II and III we get

$$N = mg \left(\frac{\sin \alpha \cos \alpha + \sin \beta \cos \beta}{\cos^2 \alpha + \cos^2 \beta} \right) = 5N \text{ Ans}$$





2.



$$mg - N - F \sin \alpha = 0 \quad [\text{Equilibrium of block in vertical direction}]$$

at breaking off the contact $N = 0$.

$$\Rightarrow F \sin \alpha = mg$$

$$\Rightarrow \text{at } \sin \alpha = \frac{mg}{F}$$

$$\Rightarrow t = \frac{mg}{a \sin \alpha}$$

$$F \cos \alpha = m A$$

[Newton's second law for block in horizontal direction]

$$\Rightarrow \text{at } \cos \alpha = m \frac{dv}{dt}$$

$$\int_0^v dv = \frac{a \cos \alpha}{m} \int_0^{t = \frac{mg}{a \sin \alpha}} t \, dt$$

$$\Rightarrow v = \frac{a \cos \alpha}{m} \frac{t^2}{2} \quad \dots\dots\dots 1$$

$$\text{After putting time limits } v = \frac{mg^2 \cos \alpha}{2a \sin^2 \alpha}$$

$$\text{equation I can be written as } \frac{dx}{dt} = \frac{a \cos \alpha}{2m} t^2$$

$$\int_0^x dx = \frac{a \cos \alpha}{2m} \int_0^t t^2 \, dt = \frac{a \cos \alpha}{2m} \frac{t^3}{3}$$

$$\text{After putting limits. } x = \frac{m^2 g^3 \cos \alpha}{6a^2 \sin^3 \alpha}$$

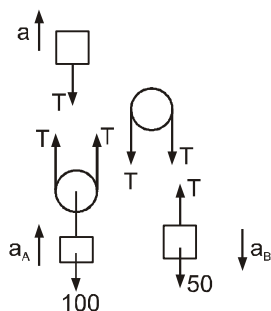
3.
$$v_{nm} = \frac{v_B + v_A/2}{2} = \frac{4 + 4/2}{2} = \frac{4 + 2}{2} = 3$$

4.
$$a_A = \frac{d^2 y}{dt^2} = \frac{1}{2}$$

$$a_B = 8a_A \quad \text{by constrained relation}$$

$$a_B = 4 \, \text{m/s}^2$$

5.





$$2a_A = a + a_B$$

$$2a_A = 3 + a_B$$

$$2T - 100 = 10a_A$$

$$50 - T = 5a_B$$

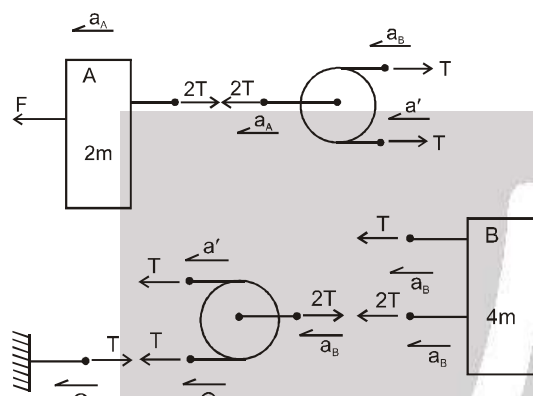
$$\Rightarrow a_B + a_A = 0$$

$$2a_A - 3 + a_A = 0$$

$$a_A = 1 \text{ m/s}^2$$

$$\Rightarrow a_B = -1 \text{ m/s}^2$$

6.



$$a_B + a' = 2a_A \quad [\text{constrained relation for pulley 1}]$$

$$O + a' = 2a_B \quad [\text{constrained relation for pulley 2}]$$

From above two equations

$$3a_B = 2a_A$$

$$\Rightarrow a_A = \frac{3}{2} a_B$$

.....I

$$F - 2T = 2ma_A \quad [\text{Newton's II law for block A}]$$

.....II

$$3T = 4m a_B \quad [\text{Newton's II law for block B}]$$

.....III

From equation I, II and III

$$a_B = \frac{3F}{17m}$$

7.

$$m_A g - 2T = m_A a_A \quad [\text{Newton's II law for block A}]$$

$$T - m_B g = m_B a_B \quad [\text{Newton's II law for block B}]$$

$$a_B + O = 2a_A \quad [\text{constrained relation for pulley P1}]$$

$$m_A = 4m_B \quad [\text{Given in question}]$$

From above four equations

$$a_A = \frac{g}{4} = 2.5 \text{ m/s}^2$$

$$a_B = \frac{g}{2} = 5 \text{ m/s}^2$$

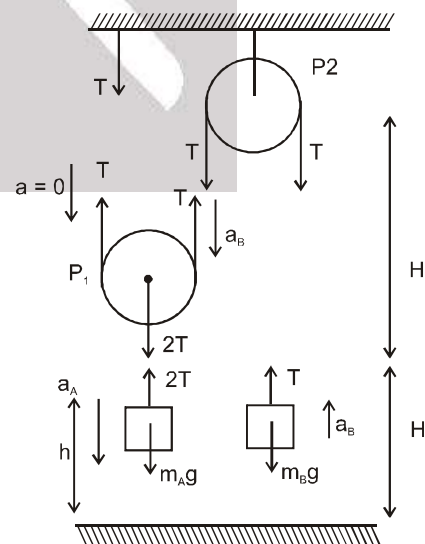
$$h = \frac{1}{2} a_A t^2 \quad [\text{Equation of motion for block A}]$$

$$\Rightarrow t = \frac{2}{5} \text{ sec.}$$

H is the distance travelled by block B in vertical direction till $\frac{2}{5}$ second

$$\Rightarrow H = \frac{1}{2} a_B t^2$$

[Equation of motion for block B]





$$\Rightarrow \frac{1}{2} 5 \left(\frac{2}{5} \right)^2$$

$$H = 0.4 \text{ m}$$

H' is the distance travelled by block B due to gained velocity.

$$v_1 = at$$

$$= 5 \times 0.4$$

$$v_1 = 2 \text{ m/s}$$

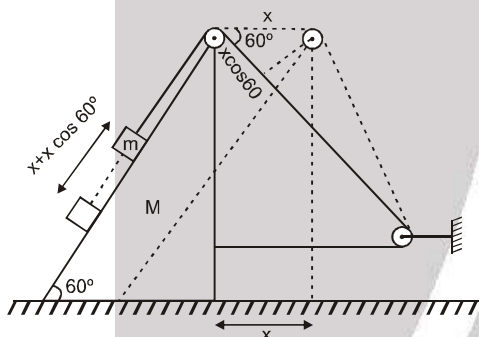
$$v_2^2 = v_1^2 + 2aH'$$

$$0^2 = 2^2 + 2(-10)H'$$

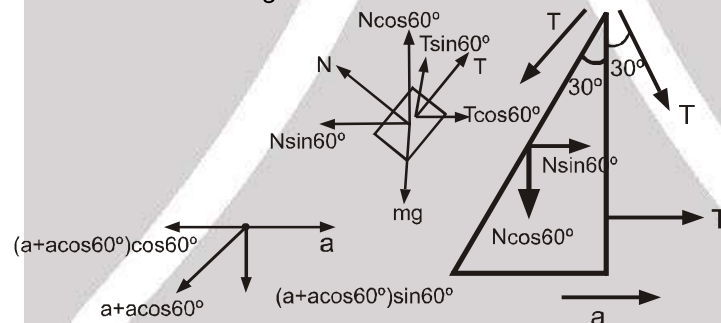
$$H' = \frac{2}{10} = 0.2 \text{ m}$$

$$\begin{aligned} \text{Total distance} &= H + H' \\ &= 0.6 \text{ m} = 60 \text{ cm.} \end{aligned}$$

8.



$\Rightarrow$ If acceleration of wedge is x then acceleration of block w.r.t. wedge is $x + x \cos 60^\circ$.



$$T + N \sin 60^\circ = Ma$$

$$T + N \frac{\sqrt{3}}{2} = Ma$$

$$T \cos 60^\circ - N \sin 60^\circ = m[a - a \cos 60^\circ - a \cos^2 60^\circ]$$

$$\frac{T}{2} - \frac{N\sqrt{3}}{2} = ma \left[1 - \frac{1}{2} - \frac{1}{4} \right]$$

$$\Rightarrow T - N\sqrt{3} = \frac{ma}{2}$$

$$mg - N \cos 60^\circ - T \sin 60^\circ = m(a \sin 60^\circ + a \cos 60^\circ \sin 60^\circ)$$

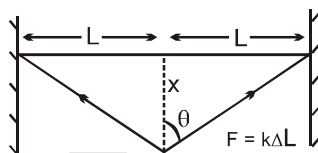
$$mg - \frac{N}{2} - \frac{T\sqrt{3}}{2} = ma \left[\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{4} \right]$$

$$2mg - N - T\sqrt{3} = \frac{3\sqrt{3}}{2} ma \Rightarrow a = \frac{30\sqrt{3}}{23} \text{ m/s}^2.$$





9.



$$\Delta L = \sqrt{L^2 + x^2} - L$$

$$F_{\text{net}} = mg - 2F \cos \theta$$

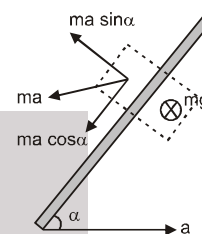
$$a_{\text{net}} = g - \frac{2k}{m} (\sqrt{L^2 + x^2} - L) \frac{x}{\sqrt{L^2 + x^2}}$$

10. Acceleration of bead along rod is

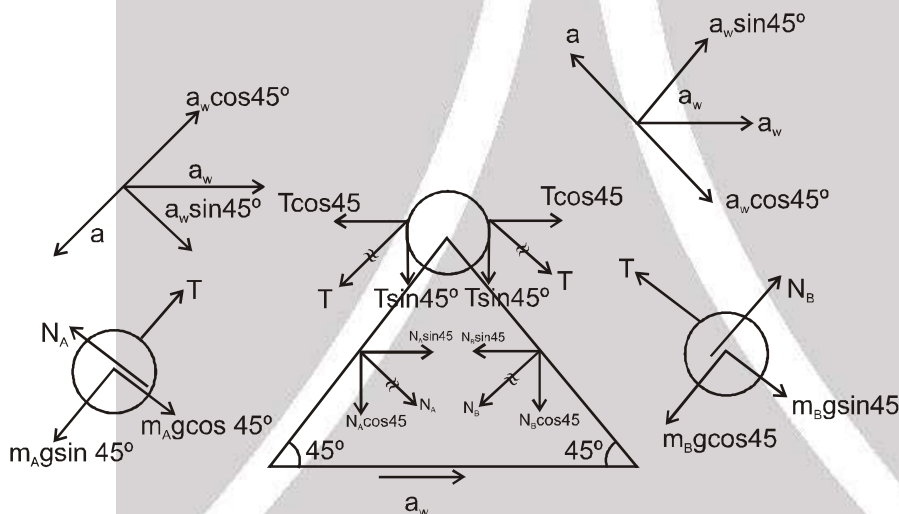
$$\frac{ma \cos \alpha}{m} = a \cos \alpha$$

$$\frac{1}{2} a \cos \alpha t^2 = \ell$$

$$t = \sqrt{\frac{2\ell}{a \cos \alpha}} = 2 \text{ sec}$$



11.



All the forces shown are in ground frame. a_w is the acceleration of wedge w.r.t ground and a is the acceleration of blocks w.r.t wedge.

$$m_A g \sin 45^\circ - T = m_A (a - a_w \cos 45^\circ)$$

[Newton's II law for block A along the wedge in ground frame]

$$m_A g \cos 45^\circ - N = m_A a_w \sin 45^\circ$$

[Newton's II law for block A in direction $\perp$ to the wedge in ground frame.]

$$T - m_B g \sin 45^\circ = m_B (a - a_w \cos 45^\circ)$$

[Newton's II law for block B along the wedge in ground frame.]

$$N_B - m_B g \cos 45^\circ = m_B (a_w \sin 45^\circ)$$

[Newton's II law for block B in direction $\perp$ to the wedge in ground frame]

$$N_A \sin 45^\circ + T \cos 45^\circ - N_B \sin 45^\circ - T \cos 45^\circ = m_w a_w$$

[Newton's II law for wedge in horizontal direction in ground frame].

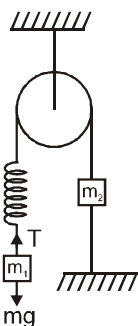
After solving above five equations we will get $a_w = \frac{2}{5} \text{ m/s}^2 = 40 \text{ cm/s}^2$





PART - III

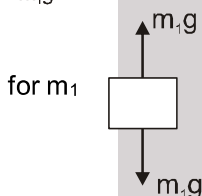
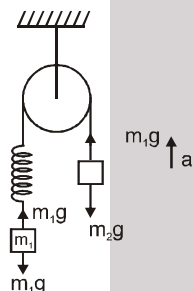
1.



$$T = m_1 g$$

when thread is burnt, tension in spring remains same = $m_1 g$.

$$m_1 g - m_2 g = m_2 a \quad \frac{(m_1 - m_2)}{m_2} g = a = \text{upwards}$$



$$a = 0$$

2.

$$F = \alpha t$$

$$a = \frac{dv}{dt} = \frac{\alpha}{m} t \quad \dots(i) \quad \text{straight line curve 1}$$

$$dv = \frac{\alpha}{m} t dt$$

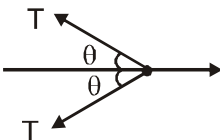
$$v = \frac{\alpha}{m} \frac{t^2}{2} \quad \text{curve 2 } \dots(ii)$$

divide (ii) by (i)

$$v = \frac{t}{2} a = \frac{a}{2} \times \frac{am}{\alpha} = \frac{a^2 m}{2\alpha}$$

→ Parabolic curve 2.

3.



$$F = 2 T \cos \theta \quad T = \frac{F}{2 \cos \theta}$$

$$\theta \uparrow \cos \theta \downarrow T \uparrow$$

on increasing θ , $\cos \theta$ decreases and hence T increases.





4. By string constraint

$$a_A = 2a_B \quad \dots\dots(1)$$

Equation for block A.

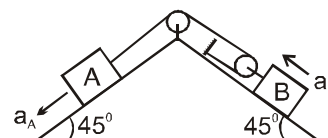
$$10 \times 10 \times \frac{1}{\sqrt{2}} - T = 10 a_A \quad \dots\dots(2)$$

Equation for block B.

$$2T - \frac{400}{\sqrt{2}} = 40 a_B \quad \dots\dots(3)$$

Solving equation (1), (2) & (3), we get $a_A = \frac{-5}{\sqrt{2}} \text{ m/s}^2$

$$a_B = \frac{-5}{2\sqrt{2}} \text{ m/s}^2 \Rightarrow T = \frac{150}{\sqrt{2}} \text{ N}$$



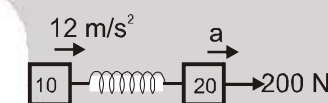
5. Apply NLM on the system

$$200 = 20 a + 12 \times 10$$

$$\frac{80}{20} = a$$

$$= 4 \text{ m/s}^2$$

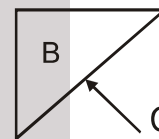
$$\text{Spring Force} = 10 \times 12 = 120 \text{ N}$$



6. There is no horizontal force on block A, therefore it does not move in x-direction, whereas there is net downward force ($mg - N$) is acting on it, making its acceleration along negative y-direction.

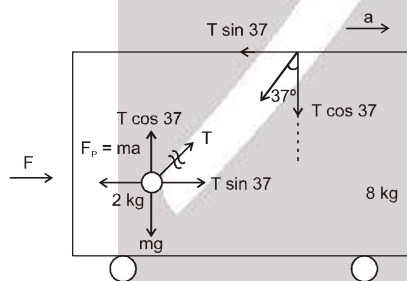
Block B moves downward as well as in negative x-direction. Downward acceleration of A and B will be equal due to constrain, thus w.r.t. B, A moves in positive x-direction.

Due to the component of normal exerted by C on B, it moves in negative x-direction.



7. Pseudo force depends on acceleration of frame and mass of object

- 8.



F.B.D. of trolley is w.r.t. ground

F.B.D. of suspended mass is w.r.t. Trolley.

$$T \cos 37^\circ - mg = 0 \quad [\text{Equilibrium of mass in y direction w.r.t. trolley}]$$

$$\Rightarrow T = \frac{5}{4} mg \quad T = 25 \text{ N}$$

$$T \sin 37^\circ - ma = 0 \quad [\text{Equilibrium of mass in x direction w.r.t. trolley}]$$

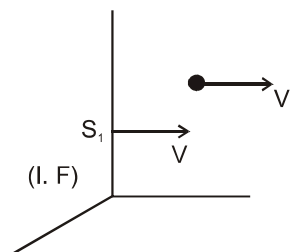
$$\Rightarrow a = \frac{T \sin 37^\circ}{m} = \frac{15}{2}$$

$$F - T \sin 37^\circ = 8a \quad [\text{Newton's II law for trolley in x direction w.r.t. ground}]$$

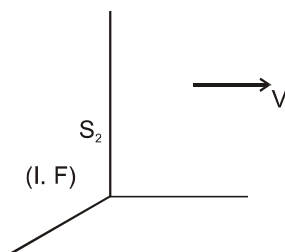
$$\Rightarrow F = 8 \times 15/2 + 25 \times 3/5 \quad F = 75 \text{ N}$$



9. (A) True
(B) True

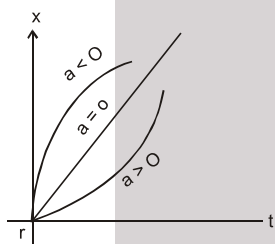


Accelerated & moving with velocity V .



Accelerated but not moving.

10.

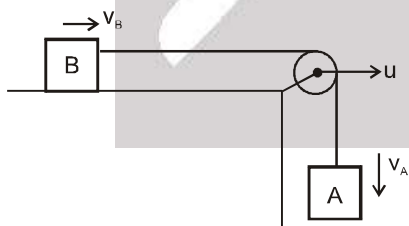


$x-t$ curve is

- (1) straight line when $a = 0$
- (2) concave up when uniform $a > 0$
- (3) concave down when uniform $a < 0$.

In the region AB & CD acceleration = 0 = Force = 0

11.



By string constrain

$$v_A + u - v_B = 0$$

or $v_B = u + v_A$

Differentiating both side

$$a_B = 0 + a_A \text{ Ans.}$$



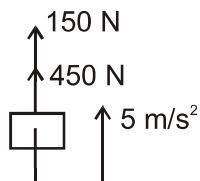


PART - IV

1. FBD of Block in ground frame :

Applying N.L. $150 + 450 - 10M = 5M$

$$\Rightarrow 15M = 600 \Rightarrow M = \frac{600}{15}$$

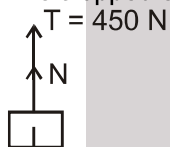


$$Mg = 10M$$

$$\Rightarrow M = 40 \text{ Kg Ans.}$$

Normal on block is the reading of weighing machine i.e. 150 N.

2. If lift is stopped & equilibrium is reached then

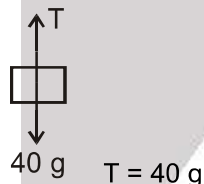


$$Mg = 400M$$

$$450 + N = 400$$

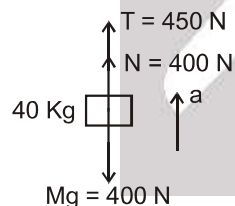
$$\Rightarrow N = -50$$

So block will lose the contact with weighing machine thus reading of weighing machine will be zero.



So reading of spring balance will be 40 Kg.

- 3.



$$a = \frac{950 - 400}{40} \Rightarrow a = \frac{450}{40} = \frac{45}{4} \text{ m/s}^2 \text{ Ans.}$$

$$4. a_p = \frac{10}{10} t = t$$

$$\therefore \frac{dv}{dt} = t \Rightarrow \int_0^v dv = \int_0^t t dt \Rightarrow v = \frac{t^2}{2}$$

Putting $v = 2$ we have $t = 2$ sec.

$$\text{Now } \frac{dx}{dt} = \frac{t^2}{2} \therefore x_p = \left[\frac{t^3}{6} \right]_0^2 = \frac{4}{3}$$

$$x_b = 2 \times 2 = 4 \text{ m}$$

$$\text{Hence relative displacement} = 4 - \frac{4}{3} = \frac{8}{3} \text{ m}$$





5. From above

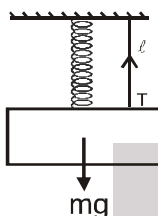
$$2t = t^3/6 \Rightarrow t^2 = 12 \Rightarrow t = 2\sqrt{3} \text{ sec.}$$

 6. $a = t = 4$
 $\therefore$ after 4 seconds $V_B = 2 \text{ m/s}$

$$V_p = 4^2/2 = 8 \text{ m/s}$$

$$\therefore V_{\text{rel}} = 8 - 2 = 6 \text{ m/s.}$$

9.



(i) $\Delta l = \ell/2$

$$F_s = K\Delta l$$

$$< \frac{2mg}{\ell} \frac{\ell}{2}$$

$$F_s < mg$$

$$T + F_s = mg$$

$$T = mg - \frac{K\ell}{2}$$

(ii) $mg - \frac{K\ell}{2} = ma$

$$g - \frac{k\ell}{2m} = a$$

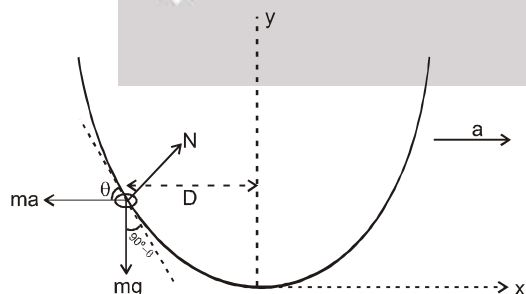
If it is so

$$F_s > mg$$

$$\text{i.e., } \Delta l < \frac{\ell}{2} \text{ string unstretched \& } T = 0.$$

EXERCISE-3 PART - I

1.



$$ma \cos \theta = mg \cos (90 - \theta)$$

$$\Rightarrow \frac{a}{g} = \tan \theta$$

$$\Rightarrow \frac{a}{g} = \frac{dy}{dx}$$

$$\Rightarrow \frac{d}{dx} (kx^2) = \frac{a}{g}$$

$$\Rightarrow x = \frac{a}{2gk} = D$$





PART - II

1. Vertical component of acceleration of A
 $a_1 = (g \sin \theta) \cdot \sin \theta$
 $= g \sin 60^\circ \cdot \sin 60^\circ = g \cdot \frac{3}{4}$

That for B

$$a_2 = g \sin 30^\circ \cdot \sin 30^\circ = g \cdot \frac{1}{4}$$

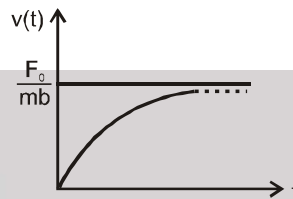
$$\therefore (a_{AB})_{\perp} = \frac{3g}{4} - \frac{g}{4} = \frac{g}{2} = 4.9 \text{ m/s}^2$$

2. $F = ma = F_0 e^{-bt}$

$$\frac{dv}{dt} = \frac{F_0}{m} e^{-bt}$$

$$\int_0^v dv = \frac{F_0}{m} \int_0^t e^{-bt} dt ; v = \frac{F_0}{m} \left[\frac{e^{-bt}}{-b} \right]_0^t$$

$$v = \frac{F_0}{mb} (1 - e^{-bt})$$



3. $a = -(g + \gamma v^2)$

$$\frac{dv}{dt} = -(g + \gamma v^2)$$

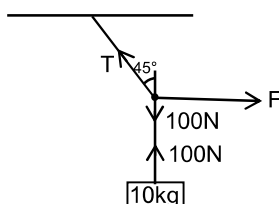
$$\int_{v_0}^0 \frac{dv}{g + \gamma v^2} = - \int_0^t dt$$

$$\frac{1}{\gamma} \int_{v_0}^0 \frac{dv}{\left(\frac{g}{\gamma} + v^2 \right)} = - \int_0^t dt$$

$$\frac{1}{\gamma} \frac{1}{\sqrt{\frac{g}{\gamma}}} \left[\tan^{-1} \left(\frac{v}{\sqrt{\frac{g}{\gamma}}} \right) \right]_{v_0}^0 = -t$$

$$\frac{1}{\sqrt{g\gamma}} \tan^{-1} \left(\frac{\sqrt{\gamma}}{\sqrt{g}} v_0 \right) = t$$

- 4.



$$\frac{T}{\sqrt{2}} = 100 ; \quad \frac{T}{\sqrt{2}} = F ; \quad F = 100\text{N}.$$

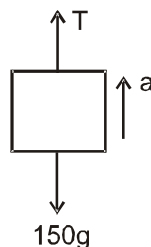




HIGH LEVEL PROBLEMS (HLP)

1. (a) (i) acceleration at $t = 1$ s

$$a = \frac{3.6 - 0}{2 - 0} = 1.8 \text{ m/s}^2$$



$$T - 150g = 150a$$

$$T = 150 \times 9.8 + 150 \times 1.8$$

$$= 1740 \text{ N}$$

(ii) At $t = 6$ s $t = 6$ s, $a = 0$

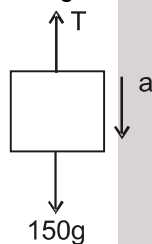
$$\therefore T = 150g \text{ N}$$

$$= 150 \times 9.8 = 1470 \text{ N}$$

(iii) At $t = 11$ s $t = 11$ s; $a = -1.8 \text{ m/s}^2$

$$1.8 \text{ m/s}^2 \text{ down}$$

$$150g - T = 150a$$



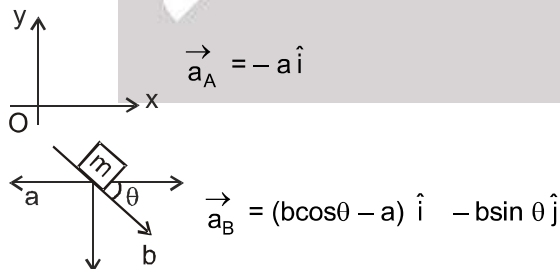
$$T = 150 \times (9.8 - 1.8) = 1200 \text{ N}$$

(b) Height = Area of $v - t$ graph
 $= \frac{1}{2}(12 + 8)3.6 = 36 \text{ m}$

(c) Average velocity = $\frac{\text{Displacement}}{\text{time}} = \frac{36}{12} = 3 \text{ m/s}$

(d) Average acceleration = $\frac{\text{change in velocity}}{\text{time in taken}} = \frac{0 - 0}{12} = 0$

- 2.



As there is no external force along x direction

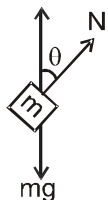
$$\therefore 2ma_{Ax} + ma_{Bx} = 0$$

$$\Rightarrow 2m(-a) + m(b \cos \theta - a) = 0$$

$$\Rightarrow 3a = b \cos \theta \dots\dots\dots (1)$$

$$\therefore \vec{a}_B = 2a \hat{i} - 3a \hat{j} \tan \theta \dots\dots\dots (2)$$





∴ along x-direction

$$N \sin \theta = m \times 2a \quad \dots\dots\dots(2)$$

Along y-direction

$$mg - N \cos \theta = m \ 3a \tan \theta \quad \dots\dots\dots(3)$$

$$\Rightarrow mg - 2ma \cot \theta = 3ma \tan \theta$$

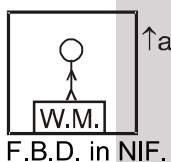
$$\Rightarrow g = a [2 \cot \theta + 3 \tan \theta]$$

$$a = \frac{g \sin \theta \cos \theta}{2 \cos^2 \theta + 3 \sin^2 \theta}$$

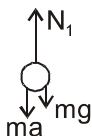
$$a = \frac{g \sin \theta \cos \theta}{3 - \cos^2 \theta}$$

$$b = \frac{3g \sin \theta}{3 - \cos^2 \theta}$$

3.

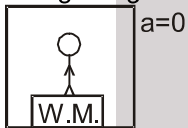


F.B.D. in N.I.F.

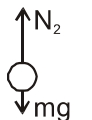


$$N_1 = mg + ma$$

$$80.5g = mg + ma \dots\dots (1)$$

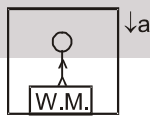
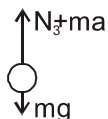


F.B.D. in



$$N_2 = mg \dots\dots\dots (2)$$

F.B.D. in N.I.F.



$$N_3 + ma = mg$$

$$\Rightarrow N_3 = mg - ma \quad \Rightarrow 59.5g = mg - ma \dots\dots\dots (3)$$

$$(1) + (3) \quad 140g = 2mg$$

m = 70 kg Ans.

(a) ∴ $N_2 = \text{true weight} = 70 \text{ kg. Ans.}$

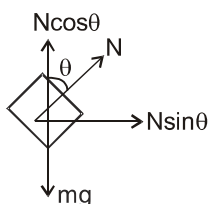
(b) by (1) $80.5 \times g = mg + ma$

$$\Rightarrow 10.5g = 70a \quad \Rightarrow a = \frac{10.5 \times 10}{70} = 1.5 \text{ m/s}^2 \text{ Ans.}$$





4. Let a be acceleration of system

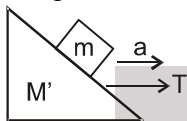


$$N \sin \theta = ma \quad \dots\dots(1)$$

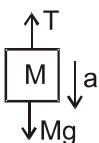
$$N \cos \theta = mg \quad \dots\dots(2)$$

Dividing (1) by (2), we get

$$a = g \tan \theta \quad \dots\dots(3)$$



$$T = (M' + m) a \quad \dots\dots(4)$$

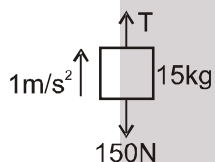


$$Mg - T = Ma \quad \dots\dots(5)$$

$$(4) + (5) \quad Mg = (M' + m + M)a \quad \dots\dots(6)$$

$$\text{by (3) \& (6)} \quad Mg = (M' + m + M)g \tan \theta \quad \Rightarrow M = \frac{M' + m}{\cot \theta - 1} \text{ Ans.}$$

- 5.



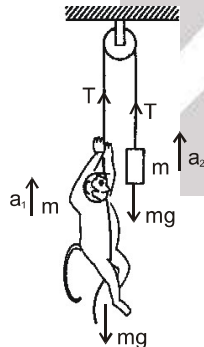
$$T - 150 = 15 \times 1$$

$$T = 165 \text{ N Ans.}$$

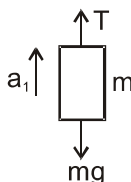
$$S = \frac{1}{2} at^2 \quad 5 = \frac{1}{2} \times 1 \times t^2$$

$$t = \sqrt{10} \text{ s Ans.}$$

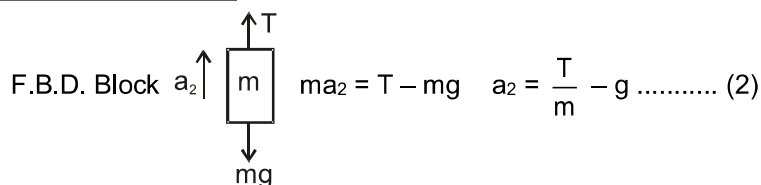
6. Let a_1 & a_2 be acceleration of monkey & Block respectively



F.B.D. Monkey



$$ma_1 = T - mg \quad a_1 = \frac{T}{m} - g \quad \dots\dots(1)$$



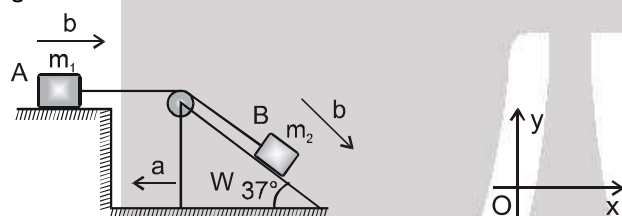
By (1) & (2)

$$a_1 = a_2 \quad \therefore \quad a_{rel} = 0, \text{ as } u_{rel} = 0$$

Relative displacement is zero.

Hence separation remains same.

7. Let b be acceleration of masses m_1 & m_2 with respect to wedge & a be acceleration of wedge w.r.t. ground.



$$\vec{a}_{WG} = -a \hat{i} \dots\dots (1)$$

$$\vec{a}_{AG} = \vec{a}_{AW} + \vec{a}_{WG}$$

$$= b \hat{i} - a \hat{i} \Rightarrow \vec{a}_{AG} = (b - a) \hat{i} \dots\dots\dots (2)$$

$$\vec{a}_{BG} = \vec{a}_{BW} + \vec{a}_{WG} = b \cos 37^\circ \hat{i} - b \sin 37^\circ \hat{j} - a \hat{i}$$

$$\vec{a}_{BG} = \left(\frac{4b}{5} - a \right) \hat{i} - \frac{3b}{5} \hat{j} \dots\dots\dots (3)$$

As $F_{\text{external}, x} = 0$

$$\Rightarrow M_A a_{AG, x} + M_B a_{BG, x} + m_W a_{WG, x} = 0$$

$$\Rightarrow 1.3(b - a) + 1.5 \left(\frac{4b}{5} - a \right) + 3.45(-a) = 0$$

$$\Rightarrow (1.3 + 1.5 + 3.45)a = (1.3 + 1.2)b$$

$$\Rightarrow 6.25a = 2.5b$$

$$\Rightarrow 5a = 2b \dots\dots\dots(1)$$

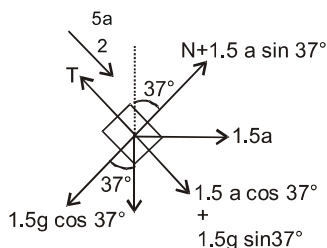
$$b - a = \frac{3}{2}a$$

F.B.D. System : m_1

Frame : $T = 1.3 \times \frac{3}{2}a \dots\dots\dots(2)$

F.B.D. System : m_2

Frame :



Along the incline :

$$1.5a \frac{4}{5} + 1.5g \frac{3}{5} - T = 1.5 \frac{5a}{2}$$

$$\Rightarrow 9 - T = 2.55 a \quad \dots\dots\dots(3)$$

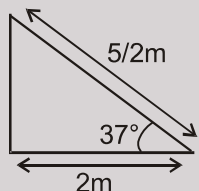
by (2) & (3) (2)

$$9 = 4.5 a$$

$$\Rightarrow a = 2 \text{ m/s}^2$$

$$\therefore b = 5 \text{ m/s}^2$$

$$S = \frac{1}{2} b t^2 \quad \Rightarrow \quad \frac{5}{2} = \frac{1}{2} \times 5 t^2 \quad t = 1 \text{ s}$$



$$(i) \quad \therefore V_{m_3} = u + a_{m_3} = 0 + 2 \times 1$$

$$\therefore V_{m_3} = 2 \text{ m/s} \quad \text{Ans.}$$

$$(ii) \quad \vec{a}_{BG} = \left(\frac{4}{5} \times 5 - 2 \right) \hat{i} - \frac{3}{5} \times 5 \hat{j}$$

$$\vec{a}_{BG} = 2 \hat{i} - 3 \hat{j}$$

$$a_{M_2} = |\vec{a}_{BG}| = \sqrt{13} \text{ m/s}^2$$

$$V_{M_2} = a_{m_2} t$$

$$V_{M_2} = \sqrt{13} \text{ m/s}^2 \quad \text{Ans.}$$

$$\text{by (2)} \quad T = \frac{3.9}{2} \times 2 \quad \Rightarrow T = 3.9 \text{ N Ans.}$$

8. $m > m'$

Let a be acceleration of M w.r.t. ground

b_1 = acceleration of m' w.r.t. ground

b_2 = acceleration of m w.r.t. ground

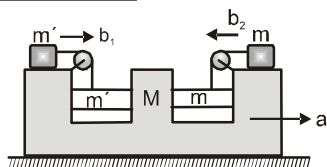
$$\vec{a}_{MG} = a \hat{i} \quad a_{m'G} = b_1 \hat{i} \quad \vec{a}_{mG} = -b_2 \hat{i}$$

As $F_{\text{external } x} = 0$

$$\Rightarrow m' a_{m'Gx} + (M + m + m') a_{MGx} + m a_{mG, x} = 0$$

$$m' b_1 + (M + m + m') a - m b_2$$

$$m' b_2 - m' b_1 = (M + m + m') a \quad \dots\dots\dots(1)$$



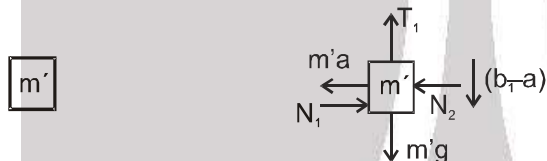
F.B.D. System : m'

Frame : I.F. $\xrightarrow{b_1}$ $T_1 = m'b_1$ (2)

F.B.D. System : M

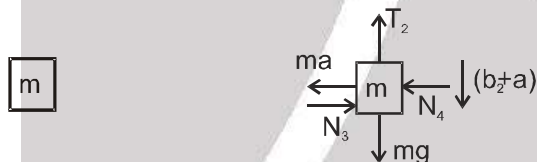
Frame : $\xrightarrow{b_2}$ $T_2 = mb_2$ (3)

F.B.D. System



Frame : N.I.F. $m'g - T_1 = m'(b_1 - a)$ (4)

F.B.D. System :



Frame : N.I.F. $mg - T_2 = m(b_2 + a)$ (5)

(2) + (4) $\Rightarrow m'g = m'(2b_1 - a)$ (6)

(3) + (5) $\Rightarrow mg = m(2b_2 + a)$ (7)

Solving (1), (6) & (7) we get

$a = \frac{(m - m')g}{2M + 3m + 3m'}$ **Ans.**

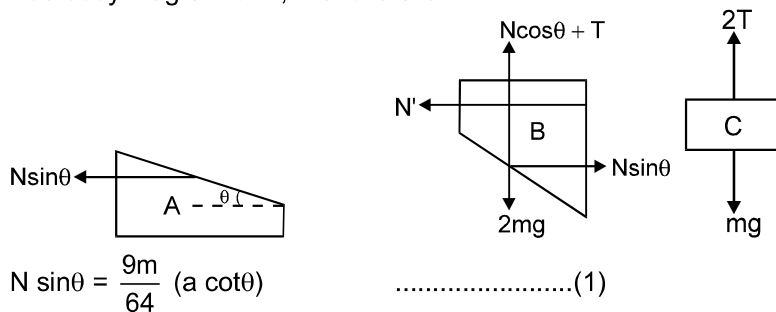
9. Let the acceleration of B downwards be $a_B = a$

From constraint ; acceleration of A and C are

$a_A = a \cot \theta = \frac{4a}{3}$ towards left

$a_C = \frac{a}{2}$ upwards

free body diagram of A, B and C are





$$2mg - T - N \cos \theta = 2ma \quad \dots\dots\dots(2)$$

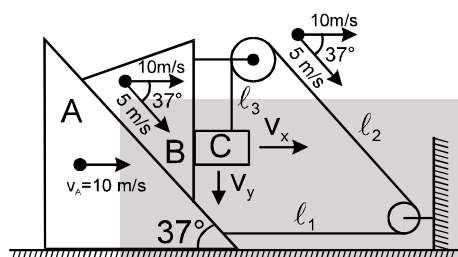
$$2T - mg = m \frac{a}{2} \quad \dots\dots\dots(3)$$

solving we get

$$a_c = \frac{a}{2} = 3\text{m/s}^2$$

Ans. 3m/s^2 upwards

10.



Let v_x and v_y be the horizontal and vertical component of velocity of block C.
The component of relative velocity of B and C normal to the surface of contact is zero.

$$\therefore 10 + 5 \cos 37^\circ - v_x = 0 \quad \dots\dots(1)$$

$$v_x = 14 \text{ m/s}$$

From the figure $l_1 + l_2 + l_3 = \text{constant}$

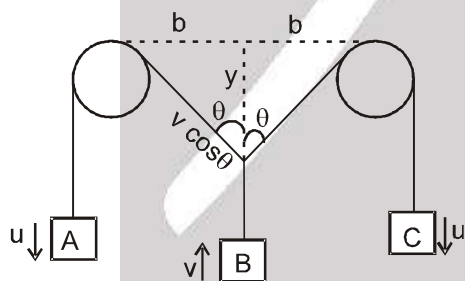
$$\therefore \frac{dl_1}{dt} + \frac{dl_2}{dt} + \frac{dl_3}{dt} = 0$$

$$(-10) + (-5 - 10 \cos 37^\circ) + (-5 \sin 37^\circ + v_y) = 0 \quad \therefore v_y = 26 \text{ m/s.}$$

11.

Pseudo force on a particle depends on mass of particle and negative acceleration of observer.

12.



$$v \cos \theta = u$$

$$v = u \sec \theta$$

$$\frac{dv}{dt} = u \sec \theta \tan \theta \frac{d\theta}{dt} \quad \dots\dots\dots I$$

$$\tan \theta = b/y$$

$$\sec^2 \theta \frac{d\theta}{dt} = -\frac{b}{y^2} \frac{dy}{dt}$$

$$= + \frac{b}{y^2} \cos^2 \theta \frac{u}{\cos \theta}$$

$$= \frac{1}{b} \frac{b^2}{y^2} \cos \theta u$$



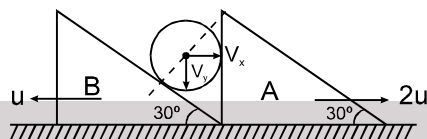
$$= \frac{u \cos \theta}{b} \tan^2 \theta \dots\dots\dots \text{II}$$

$$\Rightarrow \frac{dv}{dt} = \frac{u^2}{b} \tan^3 \theta \text{ from I and II} \quad \Rightarrow \frac{dv}{dt} = \frac{u^2}{b} \tan^3 \theta$$

13. Method - I

As cylinder will remain in contact with wedge A

$$V_x = 2u$$



As it also remain in contact with wedge B

$$u \sin 30^\circ = V_y \cos 30^\circ - V_x \sin 30^\circ$$

$$V_y = V_x \frac{\sin 30^\circ}{\cos 30^\circ} + \frac{U \sin 30^\circ}{\cos 30^\circ}$$

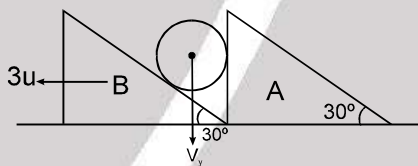
$$V_y = V_x \tan 30^\circ + u \tan 30^\circ$$

$$V_y = 3u \tan 30^\circ = \sqrt{3} u$$

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{7} u \quad \text{Ans.}$$

Method - II

In the frame of A

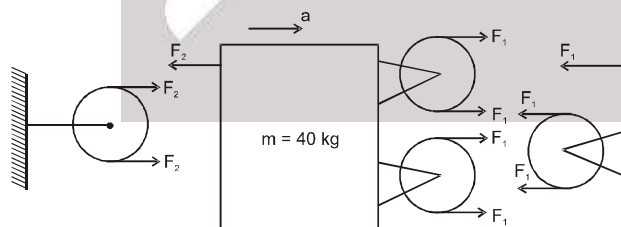


$$3u \sin 30^\circ = V_y \cos 30^\circ$$

$$\Rightarrow V_y = 3u \tan 30^\circ = \sqrt{3} u$$

$$\text{and } V_x = 2u \Rightarrow V = \sqrt{V_x^2 + V_y^2} = \sqrt{7} u \quad \text{Ans.}$$

14.



$$4F_1 - F_2 = ma \text{ [Newtons II law for block]}$$

$$\Rightarrow a = \frac{4F_1 - F_2}{m}$$

$$t = 0 \text{ to } 2 \text{ sec.}$$

$$F_1 = 30 \text{ N}$$

$$F_2 = 10 \text{ N}$$

$$\Rightarrow a = \frac{4 \times 30 - 10}{40} = 2.75 \text{ m/s}^2$$

$$t = 2 \text{ to } 4 \text{ sec}$$





$$F_1 = 30\text{N}$$

$$F_2 = 20\text{N}$$

$$\Rightarrow a = \frac{4 \times 30 - 20}{40} = 2.5 \text{ m/s}^2$$

For $t = 4$ to 6 sec.

$$F_1 = 10\text{N}$$

$$F_2 = 40\text{N}$$

$$\Rightarrow a = \frac{4 \times 10 - 40}{40} = 0 \text{ m/s}^2$$

For $t = 6$ to 12 sec

$$F_1 = 0, F_2 = 0$$

$$\Rightarrow a = 0 \text{ m/s}^2$$

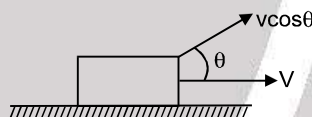
$$V_{12} - V_0 = a_{0-2}(2 - 0) + a_{2-4}(4 - 2) + a_{4-6}(6 - 4) + a_{6-12}(12 - 6)$$

$$V_{12} - 1.5 = 2.75 \times 2 + 2.5 \times 2 + 0 \times 2 + 0 \times 6$$

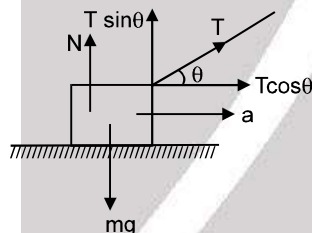
$$V_{12} = 12 \text{ m/s}$$

15. By constraint velocity component of block along the string should be u

$$\Rightarrow v \cos \theta = u \quad \text{or} \quad v = u \sec \theta \quad \dots\dots\dots(1)$$



from (1) $a = \frac{dv}{dt} = u \sec \theta \tan \theta \frac{d\theta}{dt} \quad \dots\dots\dots(2)$



Initially when block is at a large distance θ is a small component of T in vertical direction is very small. As block comes nearer and nearer. $T \sin \theta$ increases and N decreases.

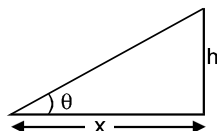
When $T \sin \theta = mg$ then block just loses contact with the ground

so $T \sin \theta = mg \quad \dots\dots\dots(3)$

$$T \cos \theta = ma \quad \dots\dots\dots(4)$$

$$(3) \text{ \& } (4) \Rightarrow$$

$$a \tan \theta = g \quad \dots\dots\dots(5)$$



also, $x = h \cot \theta$

$$\frac{dx}{dt} = -h \operatorname{cosec}^2 \theta \frac{d\theta}{dt}$$

$$\Rightarrow -v = -h \operatorname{cosec}^2 \theta \frac{d\theta}{dt} \quad [\text{as } x \text{ is decreasing } \frac{dx}{dt} = -v]$$





or $\frac{u \sec \theta}{h \csc^2 \theta} = \frac{d\theta}{dt} \dots\dots\dots(\text{using (1)}) \dots\dots\dots(6)$

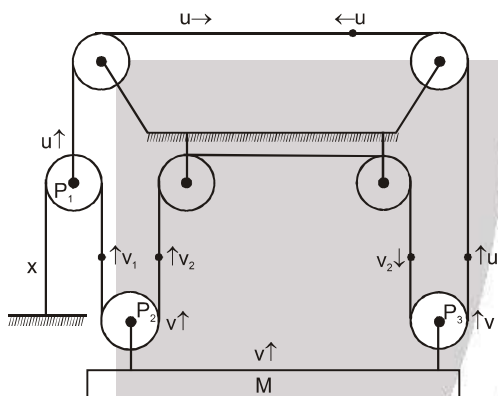
using (2) , (5) and (6) we get

$$u \sec \theta \tan \theta \left(\frac{u \sec \theta}{h \csc^2 \theta} \right) \tan \theta = g$$

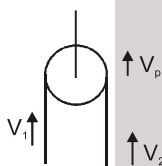
putting values of u, h & g we get.

$$\tan^4 \theta = 1 \Rightarrow \theta = \frac{\pi}{4} \quad \text{Ans.} \quad \theta = \frac{\pi}{4}$$

16.



$$V_p = \frac{V_1 + V_2}{2}$$



Pulley P₁

$$u = \frac{0 + v_1}{2} \dots\dots\dots(1)$$

Pulley P₂

$$v = \frac{v_1 + v_2}{2} \Rightarrow 2V = V_1 + V_2 \dots\dots(2)$$

Pulley P₃

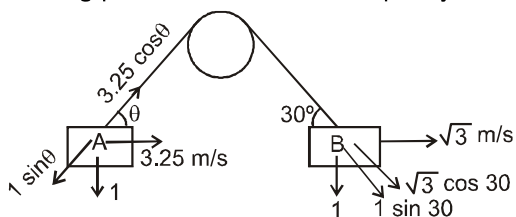
$$v = \frac{-v_2 + u}{2} \dots\dots\dots(3)$$

Eliminate V₁ & V₂ to get

$$\Rightarrow 2u + u - 2u = 2v \Rightarrow 3u = 4v$$

$$v = \frac{3}{4} u \quad \text{Ans.}$$

17. Solving problem in the frame of pulley



$$3.25 \cos \theta - 1 \sin \theta = \sqrt{3} \cos 30 + 1 \sin 30$$



$$3.25 \cos \theta - \sin \theta = \frac{3}{2} + \frac{1}{2}$$

$$3.25 \cos \theta - \sin \theta = 2$$

$$13 \cos \theta - 4 \sin \theta = 8$$

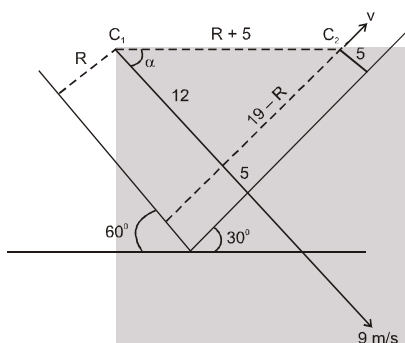
$$13\sqrt{1 - \sin^2 \theta} = 8 + 4 \sin \theta$$

$$169 - 169 \sin^2 \theta = 64 + 16 \sin^2 \theta + 64 \sin \theta$$

$$185 \sin^2 \theta + 64 \sin \theta - 105 = 0$$

$$\Rightarrow \sin \theta = \frac{3}{5} \quad \Rightarrow \tan \theta = \frac{3}{4}$$

18.



$$9 \cos \alpha = v \sin \alpha$$

$$\frac{19-R}{12} = \tan \alpha$$

$$(R+5)^2 = (12)^2 + (19-R)^2$$

$$\Rightarrow R = 10$$

Hence from (i) and (ii)

$$v = 12 \text{ m/s}$$

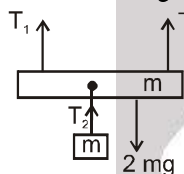
(i)

(ii)

[Pythagorean]

19.

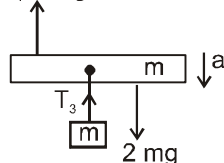
Before cutting the spring



$$T_2 = mg$$

After cutting the spring

$$T_1 = mg$$



$$2mg - mg = 2ma$$

$$a = g/2$$

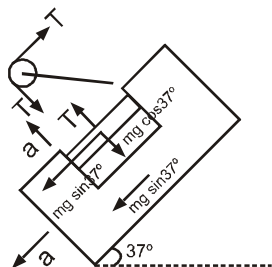
$$T_3 = mg/2$$

$$T_2 - T_3 = mg - \frac{mg}{2} = \frac{mg}{2}$$





20.



$$T - mg \cos 37^\circ = ma$$

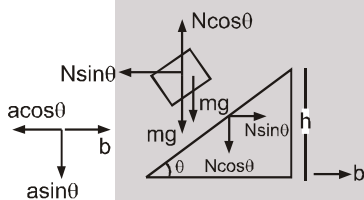
$$2mg \sin 37^\circ - T = 2ma$$

$$\Rightarrow a = \frac{4}{3} \text{ m/s}$$

$$\Rightarrow a_B = \frac{4}{3} \text{ m/s}$$

$$\Rightarrow a_A = \frac{4\sqrt{2}}{3} \text{ m/s}$$

21.



$$N \sin \theta = mb$$

$$N \sin \theta = m(a \cos \theta - b)$$

$$2mg - N \cos \theta = ma \sin \theta$$

$$\Rightarrow a = \frac{4g \sin \theta}{1 + \sin^2 \theta}$$

$$\Rightarrow h = \frac{1}{2} a \sin \theta t^2 \quad \Rightarrow \quad t = \sqrt{\frac{h(1 + \sin^2 \theta)}{2g \sin^2 \theta}}$$

